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1. Active Directory Introduction

1.1. Overview

To manage the XYZ Company network infrastructure, we installed and configured Active Directory (AD). During our classroom sessions, we used AD as the central authority to organize and secure all network resources. Instead of managing each computer or individual user manually, AD allows us to handle everything from a single point, which is the XYZ-DC1 domain controller.

In this project, the domain acts as a logical container for our departments like Marketing, Sales, and Finance. As noted in our class materials and supported by Tomsho (2017), a domain is essentially a grouping of objects that share a common directory database. One of the main advantages of this setup is how it handles data efficiency through replication. For example, when a user's password is changed, AD only replicates that specific change across the network instead of copying the entire database again. This helps keep the network fast. Everything we created in the lab, from the user accounts like JHSmith to the shared folders, follows the rules of the Active Directory schema. This schema defines exactly what information can be stored for each object in our XYZ environment.

1.2. Users

User accounts are the most basic and important objects we manage in Active Directory. In our XYZ Company project, these accounts represent the actual employees, such as John Smith or Chris Medved. Each account acts as a unique credential that tells the network who the person is and what they are allowed to do.

According to Microsoft Learn (n.d.-a), these accounts are essential for authenticating identities and authorizing access to domain resources. During the setup on XYZ-DC1, we followed a specific naming convention (first initial and last name) to keep the directory organized. We also added detailed information to these objects, such as telephone numbers and department names. This makes AD more than just a login tool. It becomes a complete company phonebook. Keeping these accounts accurate is the first step in maintaining a secure and professional network.

1.3. Groups

Managing users one by one is inefficient, so we use Groups to organize them. Instead of giving permissions to each person, we place users into groups and assign rights to the group itself. This is much faster and reduces mistakes. In this implementation, we used Security Groups to control access to files and resources.

We also followed the AGDLP strategy (Accounts, Global groups, Domain Local groups, Permissions), which is a standard method for managing permissions described by Tomsho (2017). By using groups like Marketing or Sales, we can ensure that when a new employee joins the department, we only need to add them to the right group for them to have all the access they need. During the lab, we also practiced changing group scopes to see how they behave within the domain forest

1.4. Group Policies

Group Policy is a powerful feature in Windows Server that allows us to manage the working environment of users and computers centrally. Instead of manually configuring settings on every workstation at XYZ Company, we use this service to push rules across the domain. As specified in Microsoft Learn (n.d.-c), Group Policy provides a centralized way to manage and configure operating systems, applications, and user settings in an Active Directory environment.

To apply these settings, we create Group Policy Objects (GPOs). A GPO is essentially a collection of settings that define how the system looks and behaves for a specific group of users or computers. For example, during our lab, we created GPOs to prevent unauthorized access to the Control Panel and to automatically set a professional wallpaper for the Marketing department. We also used GPOs to automate software deployment, such as installing 7-Zip on all workstations without having to run the installer manually on each one. By linking these GPOs to specific Organizational Units (OUs), we ensure that the right rules are applied to the right departments on XYZ-DC1. This granular control allows us to isolate settings for Marketing, Sales, or Finance without affecting the rest of the domain

Behind the scenes, every GPO is stored in two places on the server: the Group Policy Container, which lives inside Active Directory and keeps track of version information, and the Group Policy Template, which sits inside the SYSVOL folder and holds the actual settings. The order in which Windows applies these policies matters too. Following the LSDOU sequence (Local, Site, Domain, and finally OU), if two policies happen to conflict, the one closest to the user wins. This is exactly why our Marketing Policy linked to the Marketing OU takes precedence over anything set at the domain level and why we didn't link it at the domain root.

Group Policy is not applied once and forgotten. Computer settings refresh when the machine starts up, user settings apply at sign-in, and both refresh in the background roughly every 90 minutes (with a small random offset so all clients do not contact the Domain Controller at the same time). Security-related settings refresh at least every 16 hours regardless. During this project, instead of waiting for the next refresh cycle, we forced an immediate update using the command `gpupdate /force` from Command Prompt (Administrator Account). To manage all of this from a single interface, we used the Group Policy Management Console (GPMC), which lets us create, link, edit, and verify GPOs across the entire xyz.local domain.

2. Getting Started

This chapter covers the actual implementation steps we performed on the XYZ-DC1 server during the laboratory sessions. Every step follows the technical requirements provided by the XYZ Company Board of Trustees.

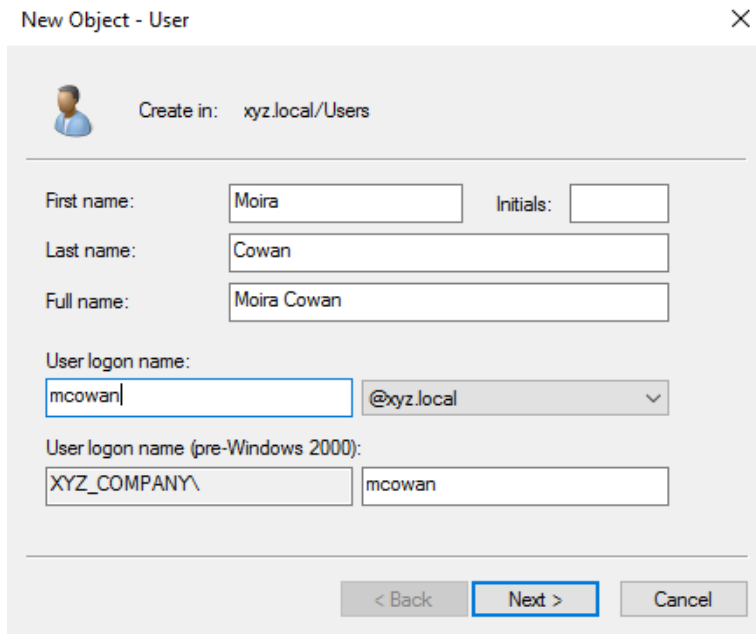
2.1. Users

The first task in our local environment was to populate the directory with the employee accounts. We focused on accuracy to ensure each user could authenticate properly within their assigned department.

2.1.1. Creating Users

To create the accounts, we used the Active Directory Users and Computers (ADUC) tool. Following the naming convention of "first initial + last name," we registered each employee. For example, when creating John Smith for the Help Desk, we set the logon name to JSmith. In cases like John Harold Smith from Marketing, we used JHSmith to prevent a conflict with the existing JSmith account.

1. Open Server Manager, click Tools, and select Active Directory Users and Computers.
2. Navigate to the XYZ.local domain and locate the specific Organizational Unit (OU) for the department.
3. Right-click the OU, select New, and then click User.
4. Fill in the First Name, Last Name, and Full Name fields as shown in the lab demonstration.
5. Enter the User logon name following the naming convention.
6. Click Next to advance to the password page. In both password fields type Password01.
7. Uncheck the option "User must change password at next logon" since the password remains stable for this project.
8. Check the option "User cannot change password" so the employee cannot modify it later. The IT department keeps centralized control over credential changes.
9. Click Next, then Finish to create the user.

A screenshot of the 'New Object - User' wizard in Active Directory. The window title is 'New Object - User' with a close button (X) in the top right. Below the title bar, there is a user icon and the text 'Create in: xyz.local/Users'. The main area contains several input fields: 'First name:' with 'Maira', 'Last name:' with 'Cowan', and 'Full name:' with 'Maira Cowan'. There is an 'Initials:' field which is empty. Below these is the 'User logon name:' section, which has a text box containing 'mcowan' and a dropdown menu showing '@xyz.local'. Underneath is the 'User logon name (pre-Windows 2000):' section, with two text boxes: the first contains 'XYZ_COMPANY\' and the second contains 'mcowan'. At the bottom of the wizard are three buttons: '< Back', 'Next >', and 'Cancel'. The 'Next >' button is highlighted with a blue border.

New Object - User

Create in: xyz.local/Users

First name: Maira Initials:

Last name: Cowan

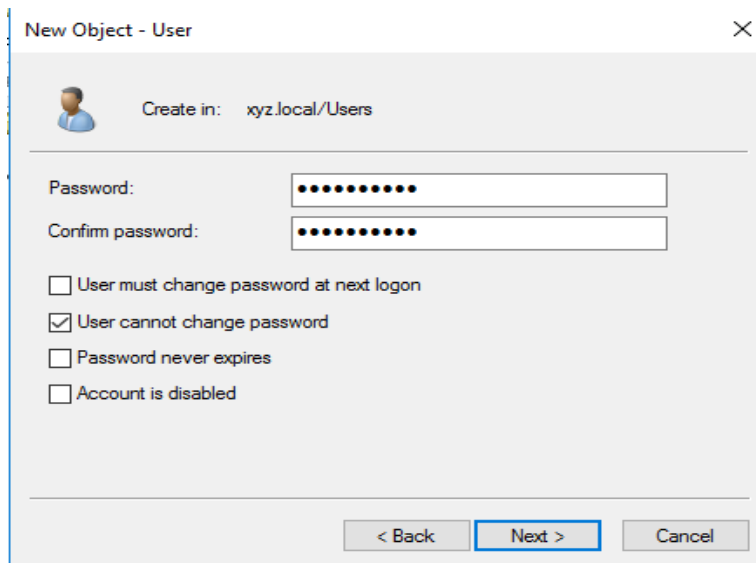
Full name: Maira Cowan

User logon name: mcowan @xyz.local

User logon name (pre-Windows 2000): XYZ_COMPANY\ mcowan

< Back Next > Cancel

Figure: New User wizard. The “Create in:” line shows the destination OU.

A screenshot of the 'New Object - User' wizard, showing the password configuration step. The window title is 'New Object - User' with a close button (X) in the top right. Below the title bar, there is a user icon and the text 'Create in: xyz.local/Users'. The main area contains two password input fields: 'Password:' and 'Confirm password:', both filled with black dots. Below these fields are four checkboxes: 'User must change password at next logon' (unchecked), 'User cannot change password' (checked), 'Password never expires' (unchecked), and 'Account is disabled' (unchecked). At the bottom of the wizard are three buttons: '< Back', 'Next >', and 'Cancel'. The 'Next >' button is highlighted with a blue border.

New Object - User

Create in: xyz.local/Users

Password:

Confirm password:

☐ User must change password at next logon

☒ User cannot change password

☐ Password never expires

☐ Account is disabled

< Back Next > Cancel

Figure: Setting the initial password and the User cannot change password flag.

2.1.2. Adding Users Information

After creating the basic accounts, we must populate them with specific organizational details. This process transforms Active Directory from a simple list of names into a robust, centralized directory for the entire company. Follow these steps to update user profiles:

1. In Active Directory Users and Computers, navigate to the Marketing OU and double-click on Moira Cowan to open the user Properties.
2. On the General tab, enter the Description (Marketing Executive), Office (Toronto), Telephone number ((416) 555-0155), E-mail (mcowan@xyz.com), and Web page (<http://www.xyz.com>).
3. Click the Address tab and fill in Street (123 Alphabet Ave.), City (Toronto), State/Province (Ontario), and Country/Region (Canada).
4. Click the Organization tab and enter Job Title (VP of Marketing), Department (Marketing), and Company (XYZ Company).

Configuring Logon Hours restrictions:

The XYZ Company security policy requires that employees only be able to log on during business hours. We configured this restriction per user account from the Account tab of the user properties dialog.

1. In Active Directory Users and Computers, double-click the user account (for example, Moira Cowan) to open the Properties window.
2. Click the Account tab.
3. Click the Logon Hours button to open the weekly schedule grid.
4. To start from a clean state, click the All button (top-left corner) to select the entire week, then choose Logon Denied. The grid will appear empty (white).
5. Click and drag from Monday 7:00 AM to Friday 8:00 PM to select the work-hour block, then choose Logon Permitted. The selected cells turn blue.
6. Click OK to close the Logon Hours dialog, then Apply and OK to save the changes to the user account.
7. We applied the same Logon Hours pattern to every user account in the project, ensuring consistency across departments.

Moira Cowan Properties

Remote control Remote Desktop Services Profile COM+
Member Of Dial-in Environment Sessions
General Address Account Profile Telephones Organization

Moira Cowan

First name: Initials:

Last name:

Display name:

Description:

Office:

Telephone number: Other...

E-mail:

Web page: Other...

OK Cancel Apply Help

Figure: General tab — display name, description, office, telephone, e-mail, and web page.

Moira Cowan Properties

Remote control Remote Desktop Services Profile COM+
Member Of Dial-in Environment Sessions
General Address Account Profile Telephones Organization

Street:

P.O. Box:

City:

State/province:

Zip/Postal Code:

Country/region:

OK Cancel Apply Help

Figure: Address tab — street, city, state, ZIP, and country.

Moira Cowan Properties

Remote control Remote Desktop Services Profile COM+
Member Of Dial-in Environment Sessions
General Address Account Profile Telephones Organization

Job Title: Vice President of Marketing

Department: Marketing

Company: XYZ Company

Manager
Name:
Change... Properties Clear

Direct reports:

OK Cancel Apply Help

Figure: Organization tab — job title, department, and company.

Logon Hours are set graphically in a 24x7 grid where blue cells mean Logon Permitted and white cells mean Logon Denied. The same process is applied to every user (a sample for Moira Cowan is shown below).

Logon Hours for Moira Cowan

12 2 4 6 8 10 12 2 4 6 8 10 12

All Sunday Monday Tuesday Wednesday Thursday Friday Saturday

OK Cancel

Logon Permitted
Logon Denied

Sunday from 12:00 AM to 1:00 AM

Figure: Logon Hours dialog — Monday through Friday 7:00 AM to 8:00 PM.

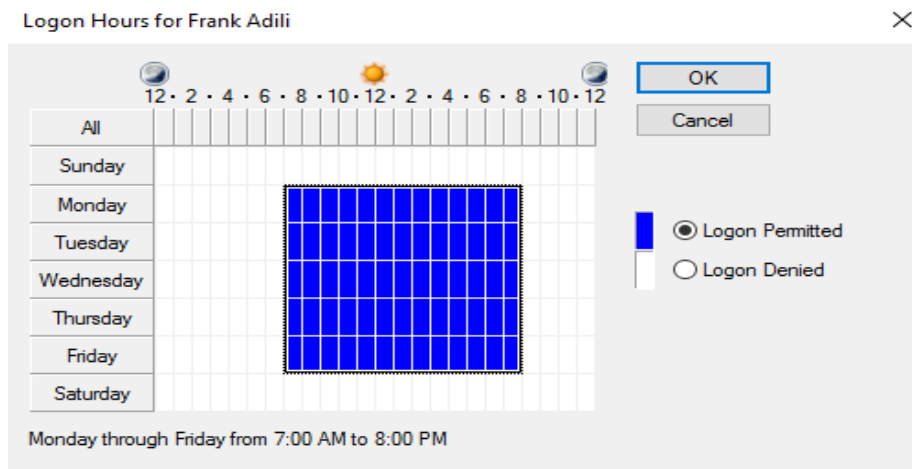


Figure: Same Logon Hours pattern applied to Frank Adili.

Once all nine users are created and their attributes (and Logon Hours) are set, the Users default container shows the complete list of accounts and security groups for the domain.

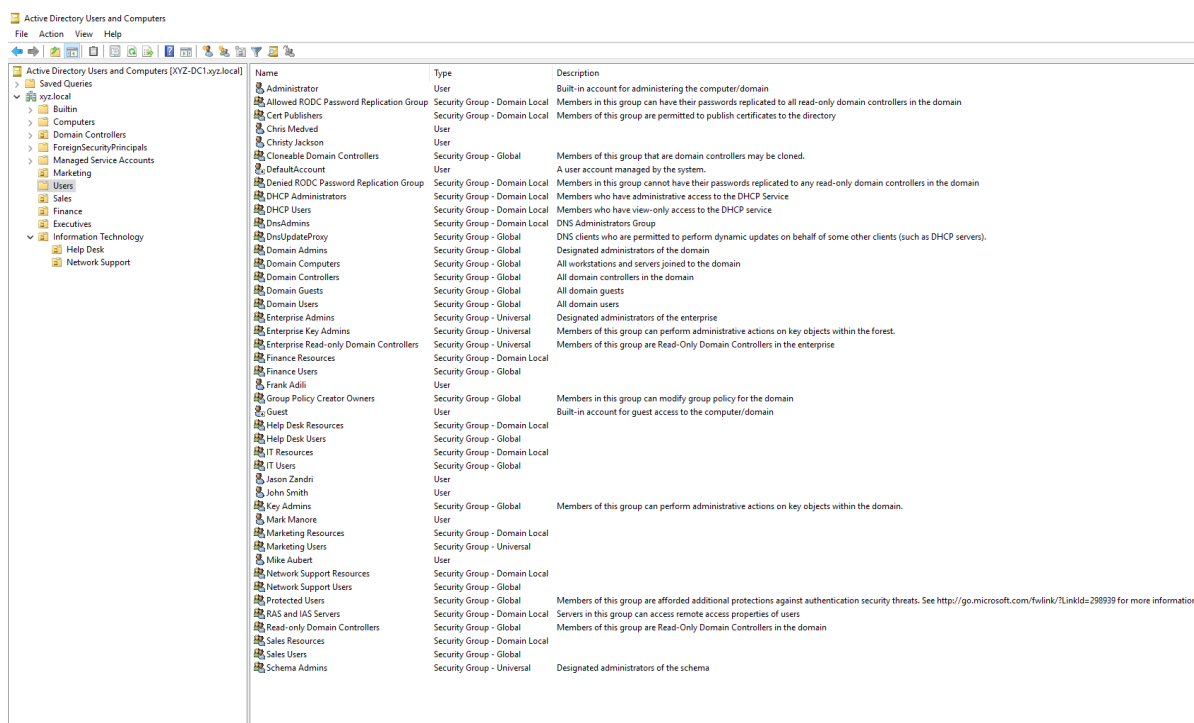


Figure: Active Directory Users and Computers — all nine users and the security groups.

2.2. Groups

Once the users were ready, the next task was to organize them into groups. This makes it much easier to manage permissions later on, especially when we start dealing with file sharing and security policies.

2.2.1. Creating Groups

For this project, we created two security groups per department. Following the specifications from the IT Department, we used the naming convention of the department name followed by "Users" and "Resources."

1. In Active Directory Users and Computers, right-click the Users folder (or the specific OU if preferred), select New, and then click Group.
2. Type the Group name exactly as shown in the table (for example, Marketing Users or Marketing Resources).
3. Set the Group scope according to the table. Marketing Users uses Universal, the other Users groups use Global, and all Resources groups use Domain Local. Leave the Group type as Security.
4. Click OK to finish.
5. Double-click the new group to open Properties, click the Members tab, click Add, type each member listed in the table, click Check Names, and click OK to add them.
6. For every Resources group, repeat step 5 but add the matching Users group as the member. This nests the Users group inside the Resources group and completes the AGDLP pattern.

The full table of groups created for the demonstration is reproduced below. After every group is created, its members are added through the Members tab of the group properties dialog.

Group	Scope	Members
Marketing Users	Universal	Moira Cowan, John Harold Smith
Marketing Resources	Domain Local	Marketing Users
Sales Users	Global	Frank Adili, Christy Jackson
Sales Resources	Domain Local	Sales Users
Finance Users	Global	Chris Medved
Finance Resources	Domain Local	Finance Users
IT Users	Global	Mark Manore
IT Resources	Domain Local	IT Users
Help Desk Users	Global	John Smith
Help Desk Resources	Domain Local	Help Desk Users
Network Support Users	Global	Mike Aubert, Jason Zandri
Network Support Resources	Domain Local	Network Support Users
Print Operators (Built-in)	Domain Local	Help Desk Users, IT Users, Network Support Users

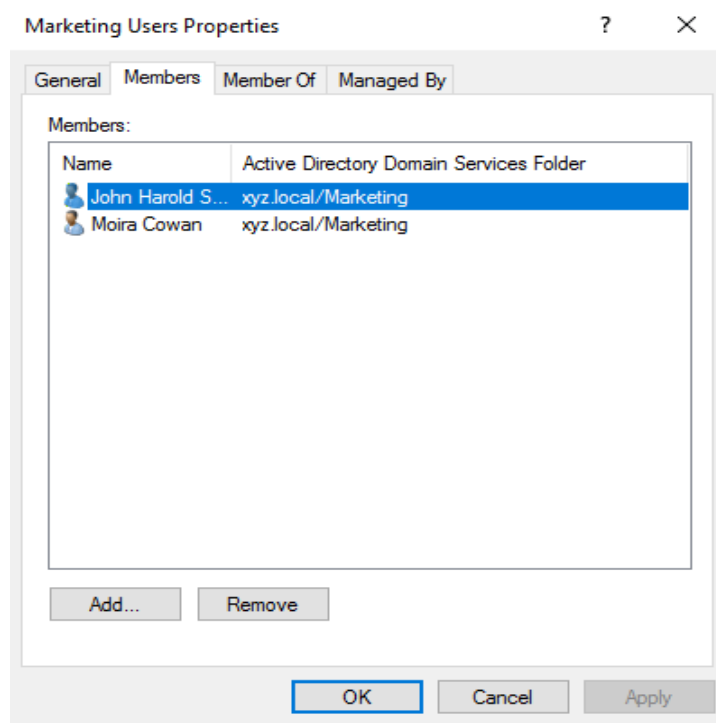


Figure: Marketing Users — Members tab showing Moira Cowan and John Harold Smith.

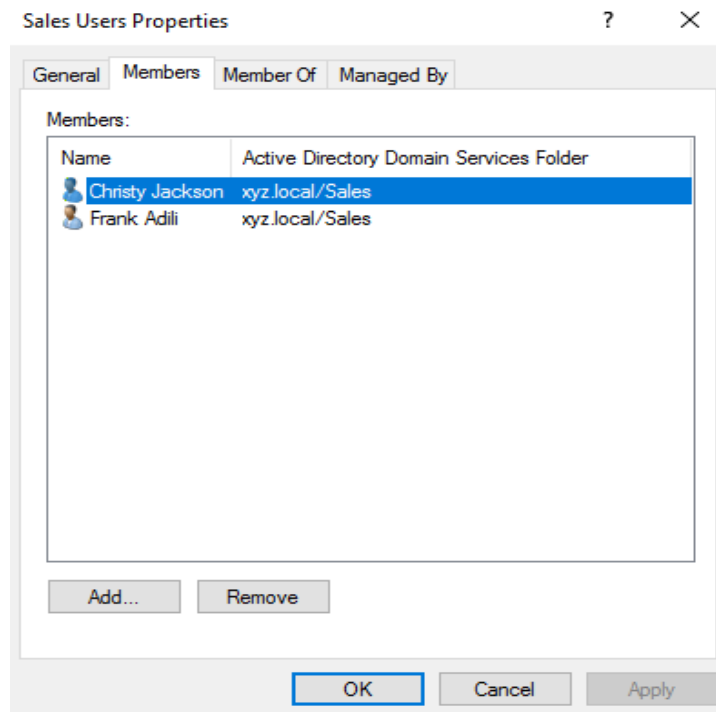
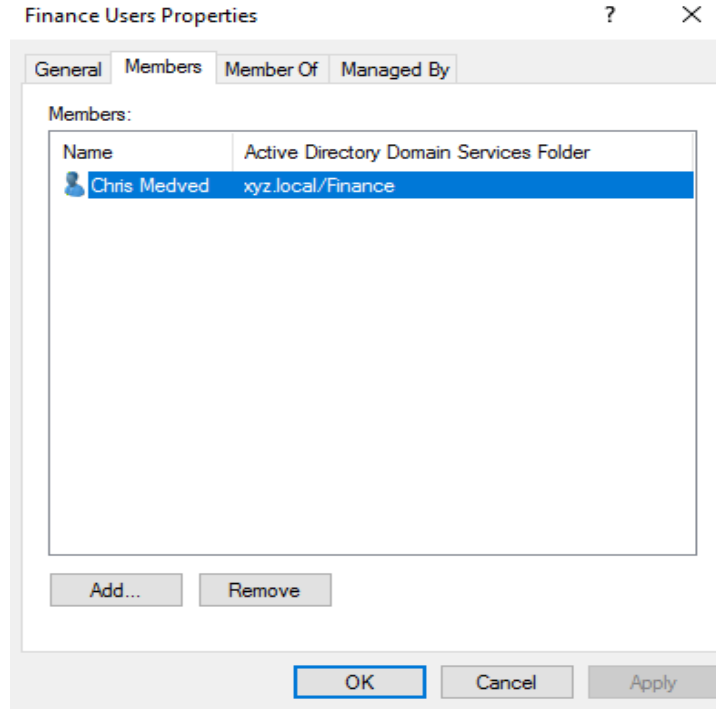
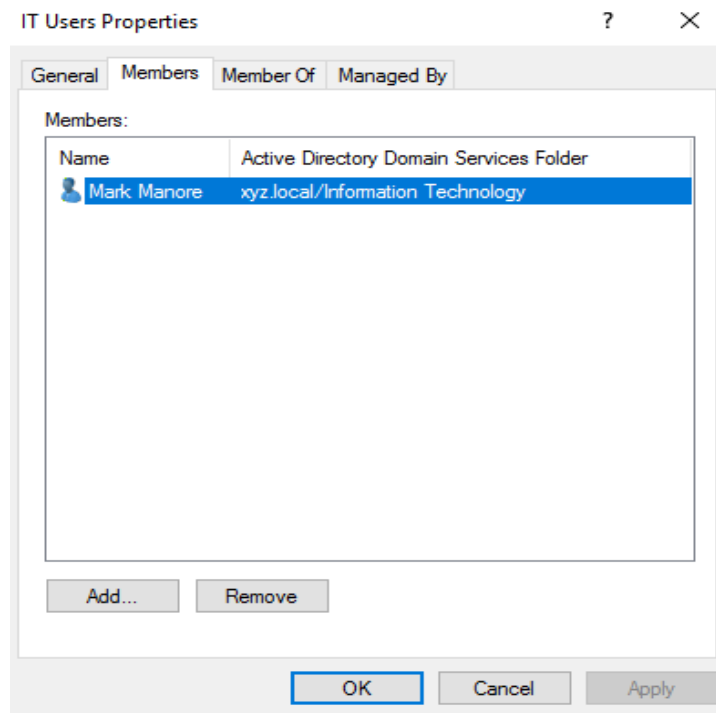


Figure: Sales Users — Frank Adili and Christy Jackson.*Figure: Finance Users — Chris Medved.**Figure: IT Users — Mark Manore.*

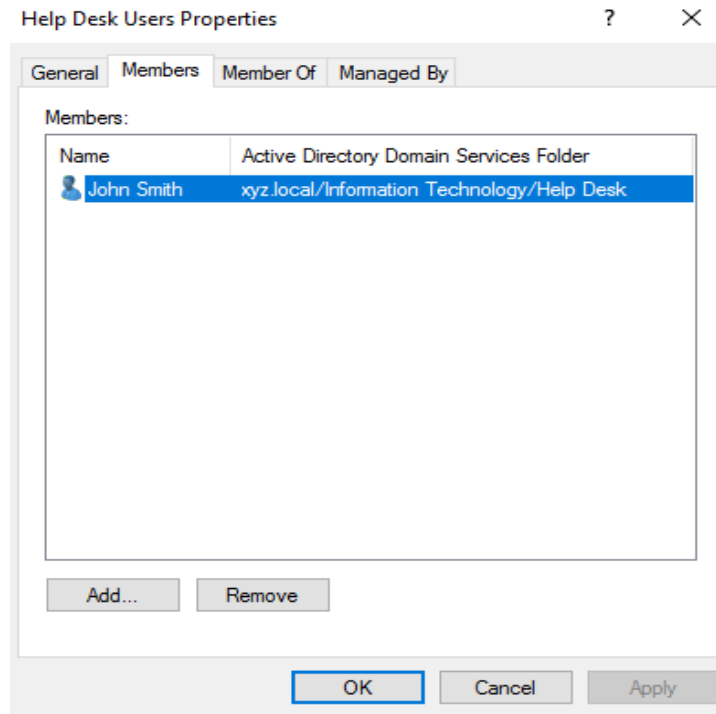


Figure: Help Desk Users — John Smith.

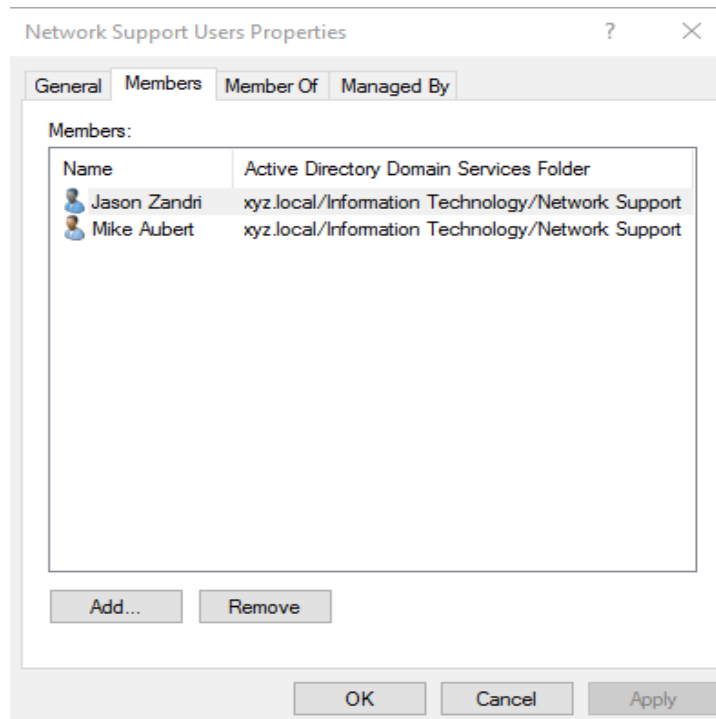


Figure: Network Support Users — Mike Aubert and Jason Zandri.

After all Users groups are filled correctly, make sure each one is added as a member of the corresponding Resources group. This is the A-G-DL-P pattern: User Accounts go into Global groups, Global groups go into Domain Local groups, permissions on resources are then granted to the Domain Local groups.

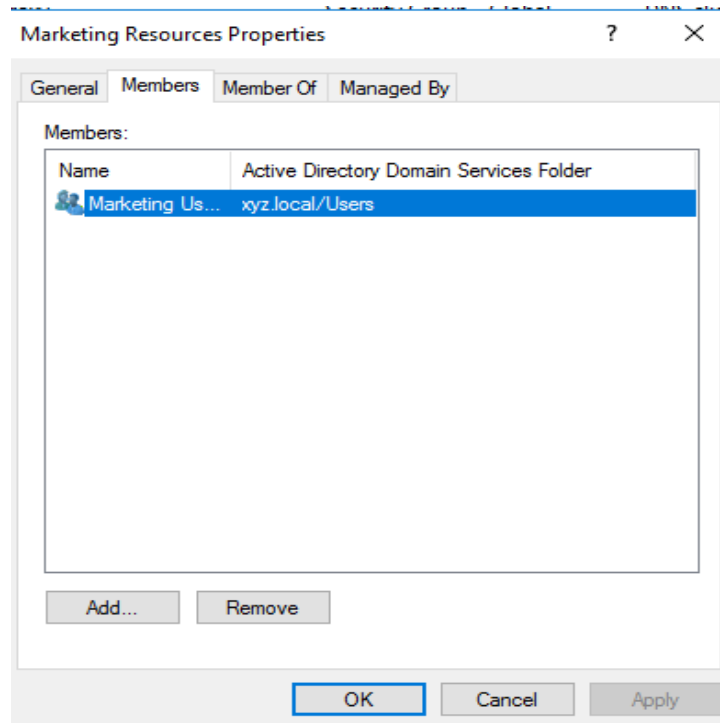


Figure: Marketing Resources contains the Marketing Users group.

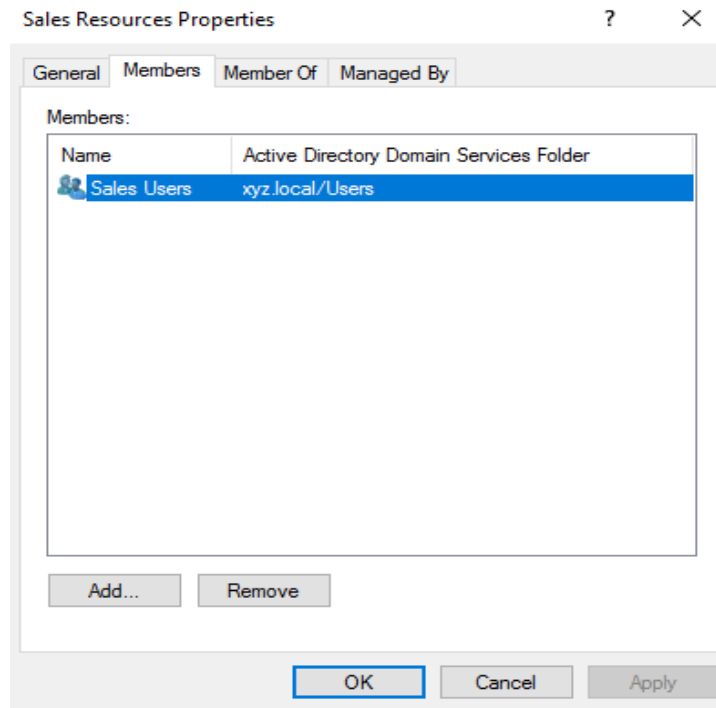


Figure: Sales Resources contains Sales Users.

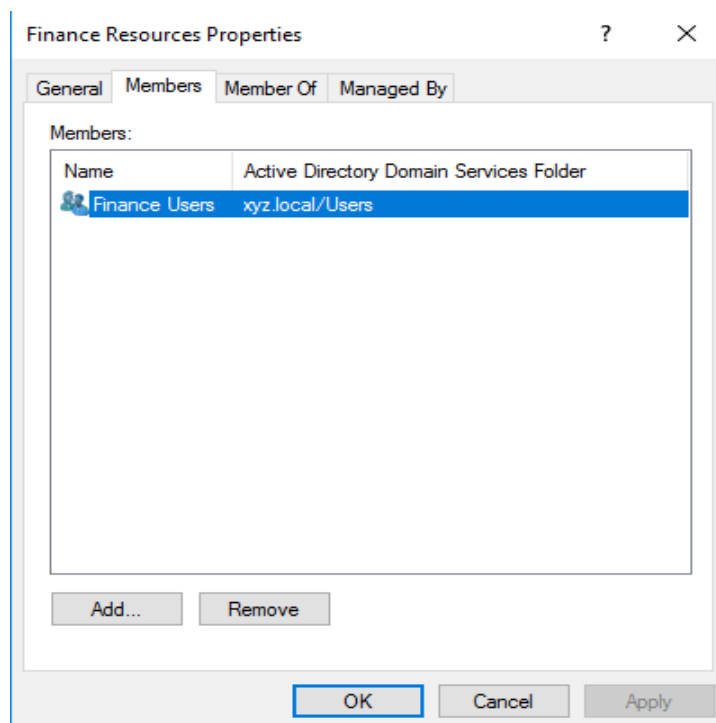


Figure: Finance Resources contains Finance Users.

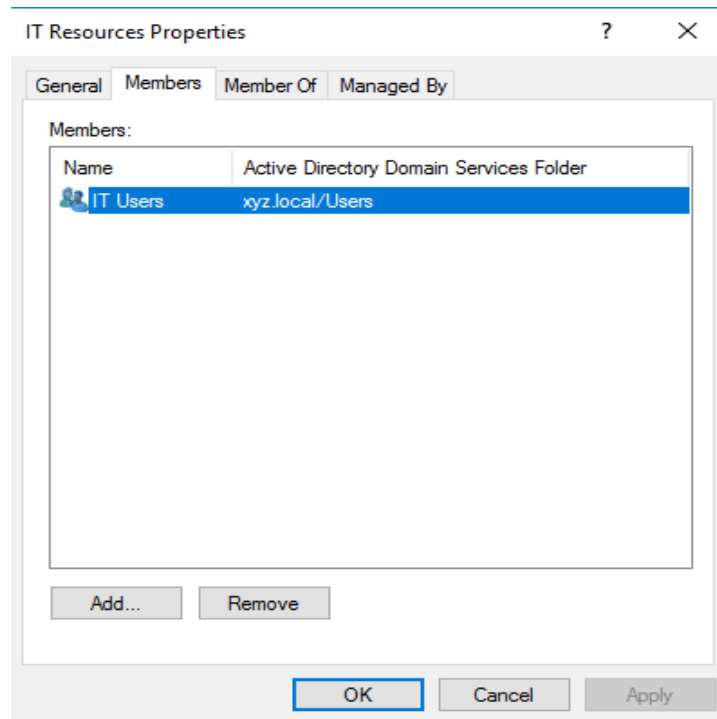


Figure: IT Resources contains IT Users.

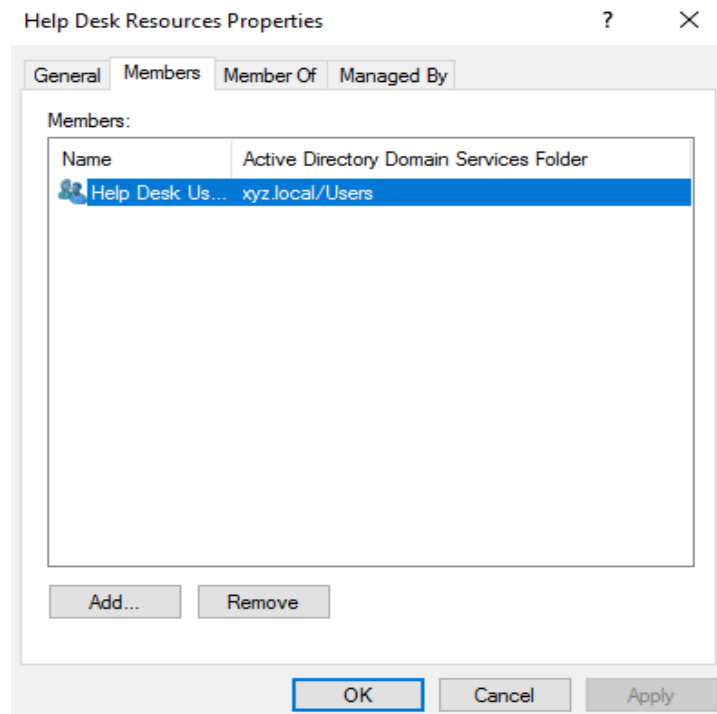


Figure: Help Desk Resources contains Help Desk Users.

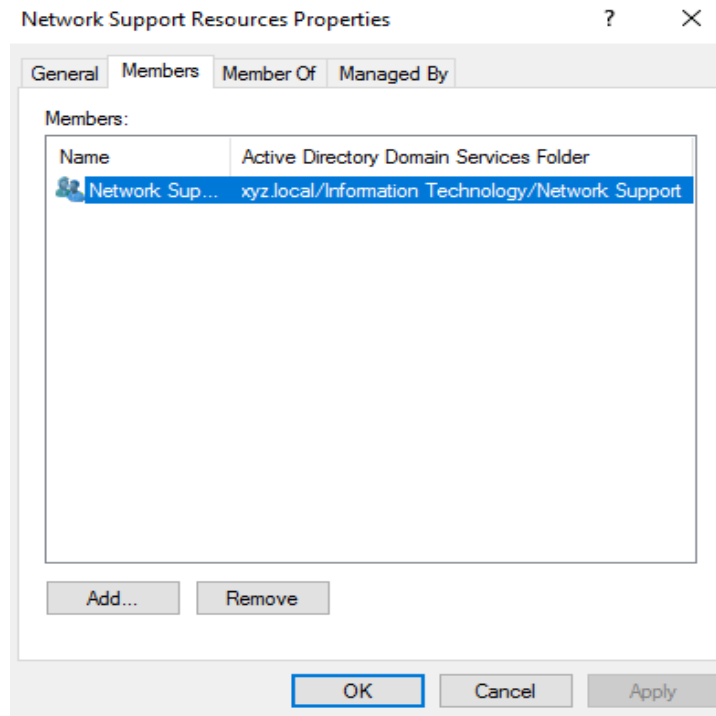


Figure: Network Support Resources contains Network Support Users.

Finally, the Help Desk Users, IT Users, and Network Support Users groups are added as members of the built-in Print Operators group. Print Operators is one of the few groups where members are allowed to log on locally to a Domain Controller, this means that the technical support staff are able to manage the printers attached to the server.

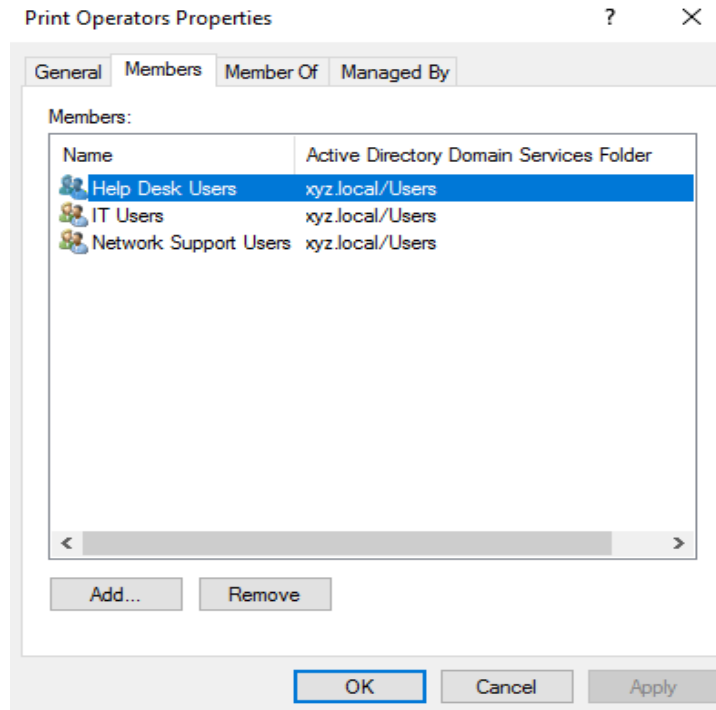


Figure: Print Operators — members from the three technical-support Users groups.

2.2.2. Changing Groups Scope

In some cases, we need to change how a group is recognized within the forest. During our class demonstration, we practiced changing a group from Global to Domain Local to understand how permissions are nested under the AGDLP strategy.

1. Right-click the group you just created (e.g., Sales Group) and select Properties.
2. Under the General tab, look at the Group scope section.
3. Change the selection from Global to Domain Local.
4. Click Apply. Note that AD allows this change as long as the group is not a member of another group that would cause a conflict in the hierarchy.

Note: Active Directory restricts scope changes based on the group's current membership rules. For example, a Global group can only be converted to a Universal scope if it is not currently a member of another Global group. This secures a structural integrity within the XYZ-DC1 database.

2.3. DNS, WINS, and DHCP

A professional network requires more than just user accounts; it needs robust name resolution and automatic IP addressing. In this section, we configure the core network services on XYZ-DC1 to ensure all devices can communicate.

2.3.1. DNS

Domain Name System (DNS) is what translates human-friendly names into IP addresses. Without it, Active Directory would not function because the client computers wouldn't be able to find the domain controller.

2.3.1.1. Creating DNS zones

1. Open Server Manager, click Tools, and select DNS.
2. In the DNS Manager, expand the XYZ-DC1 server icon.
3. Right-click Reverse Lookup Zones and select New Zone to launch the wizard.
4. On the Welcome screen, click Next.
5. Select Primary zone and check the box Store the zone in Active Directory. Click Next.
6. For the Replication Scope, choose To all DNS servers running on domain controllers in this domain: xyz.local. Click Next.
7. Select IPv4 Reverse Lookup Zone and click Next.
8. In the Network ID field, type 10 and click Next. This creates the reverse zone for the entire 10.0.0.0/8 network used by the project.
9. In the Dynamic Update screen, select Allow only secure dynamic updates. Click Next.
10. Review the summary and click Finish to create the zone.

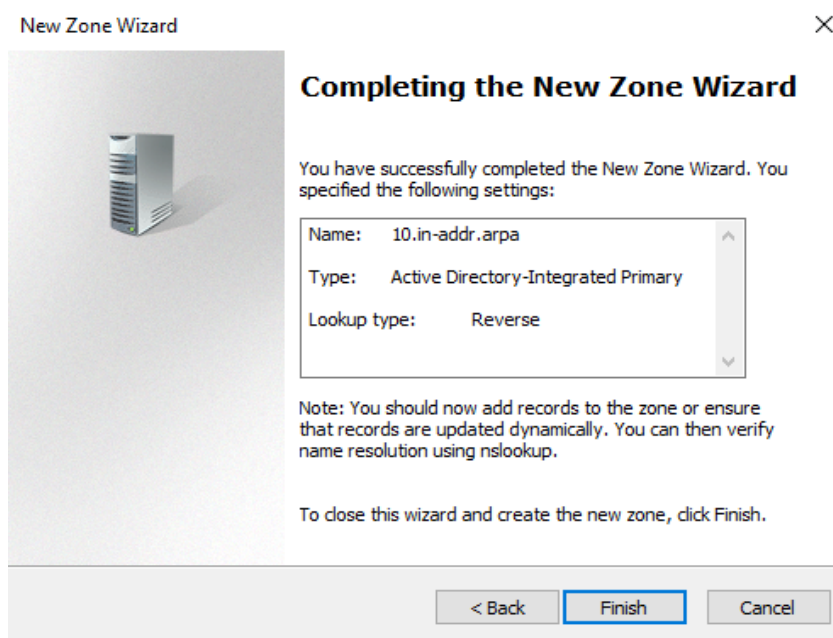
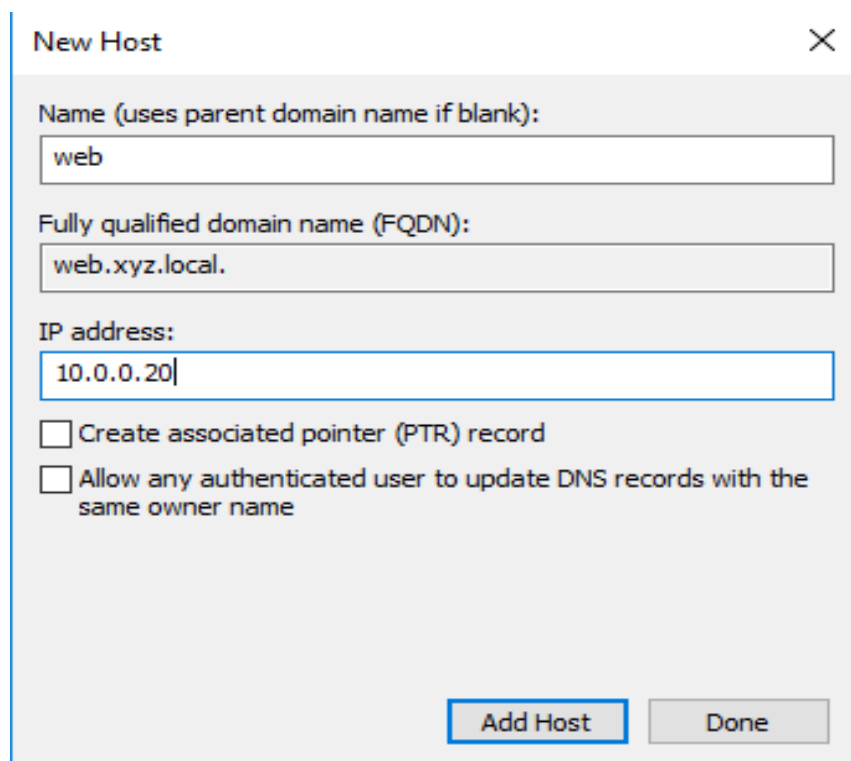


Figure: Reverse zone wizard — 10.in-addr.arpa, AD-integrated, Primary, Reverse.

2.3.1.2. Creating DNS record

Once the forward lookup zone is established, we need to add specific records so the network can locate resources by their names. In this case, we focused on Host (A) records, this maps a hostname to an IPv4 address. We also ensured that a Pointer (PTR) record was created to allow for reverse lookups, which is a common requirement for network security and troubleshooting.

1. In the DNS Manager, expand the Forward Lookup Zones folder and select the xyz.local zone we created earlier.
2. Right-click an empty area in the right pane and select New Host (A or AAAA).
3. In the Name field, type the name of the resource (for example, XYZ-DC1).
4. In the IP address field, type the static IP assigned to the server, such as 10.0.0.20.
5. Check the box labeled Create associated pointer (PTR) record.
6. Click Add Host. A confirmation message will appear stating that the host record was successfully created. Click OK.
7. If you need to add more records for other servers or printers in the XYZ Company, repeat these steps for each device.
8. Click Done to close the window and verify the new record appears in the zone list.



New Host

Name (uses parent domain name if blank):
web

Fully qualified domain name (FQDN):
web.xyz.local.

IP address:
10.0.0.20

☐ Create associated pointer (PTR) record

☐ Allow any authenticated user to update DNS records with the same owner name

Add Host Done

Figure: New Host dialog — web.xyz.local pointing to 10.0.0.20.

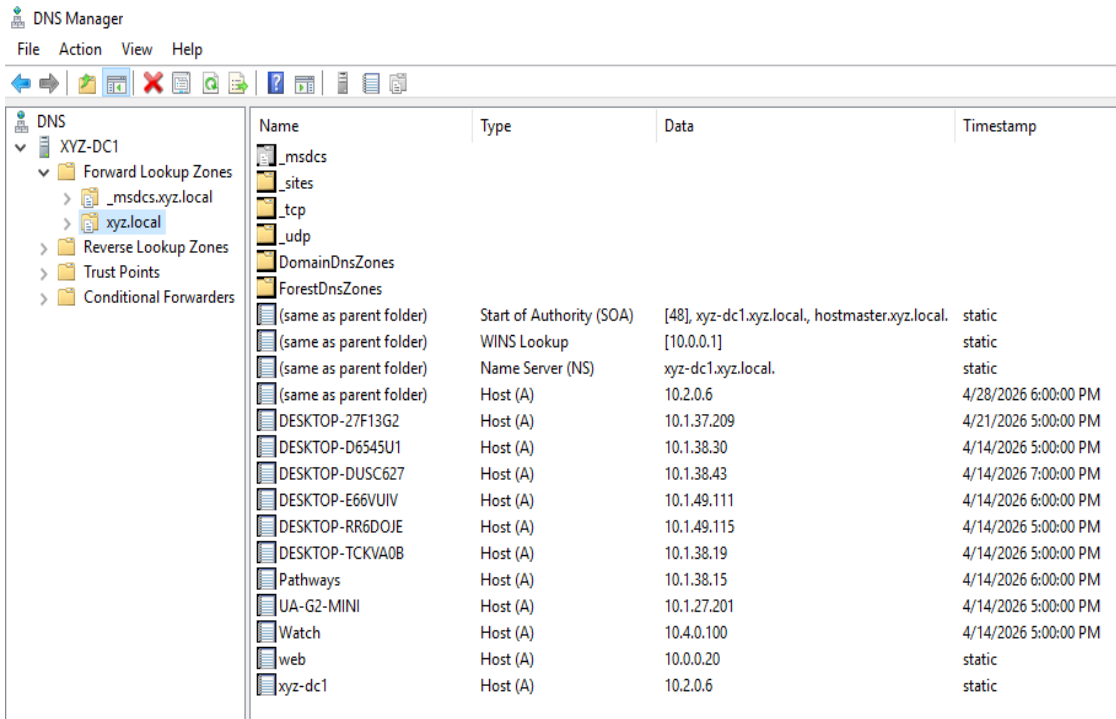


Figure: Forward zone xyz.local with all records, including the new web host and the WINS Lookup record.

2.3.2. WINS

Windows Internet Name Service (WINS) is essential for resolving NetBIOS names to IP addresses, especially in environments with legacy applications. In our XYZ Company setup, we ensure that name resolution is consistent across all protocols.

2.3.2.1. Configuring DNS

1. Open Server Manager, click Tools, and select WINS.
2. In the WINS console, expand the XYZ-DC1 server node.
3. Right-click Active Registrations and select Display Records.
4. Click Find Now to verify that the server is successfully capturing NetBIOS names from the local network.
5. Confirm that the server XYZ-DC1 [10.2.0.6] appears as Connected in the right pane of the WINS console. This confirms that WINS is running and ready to accept NetBIOS name registrations from clients on the network.

For this demonstration, we focused on confirming that XYZ-DC1 handles all local queries.

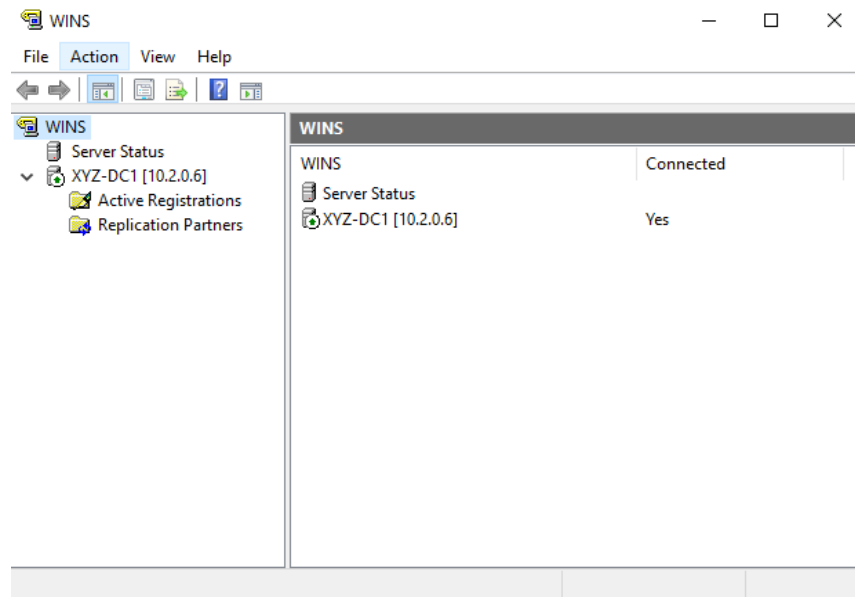


Figure: WINS console — XYZ-DC1 [10.2.0.6] connected.

Inside the DNS zone xyz.local, a WINS Lookup record is created on its own that let DNS query WINS as a fallback when a name can't be found in the zone. In addition, this integration is visible in the Forward Lookup Zones screenshot above.

The DHCP server is also configured to update DNS records dynamically on behalf of older clients that aren't able to register themselves. The dynamic update settings are accessed from the IPv4 Properties of the DHCP server, on the DNS tab.

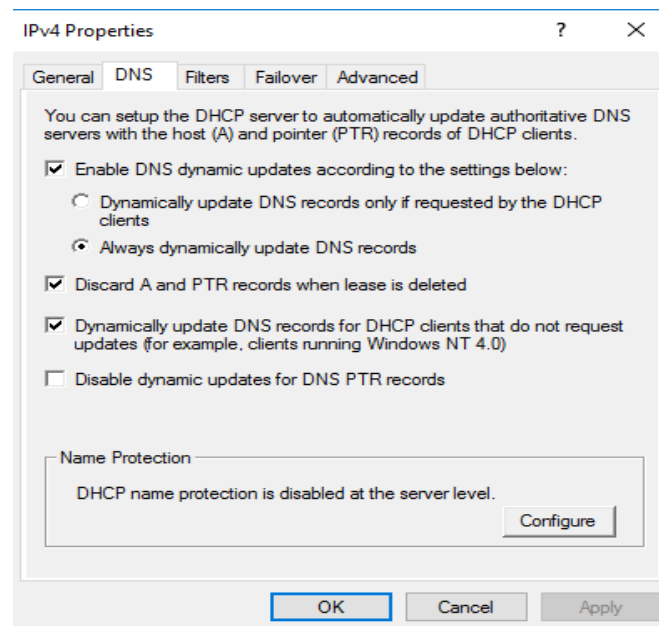


Figure: Dynamic DNS update settings on the DHCP server.

2.3.3. DHCP

The Dynamic Host Configuration Protocol (DHCP) is what allows us to automate IP management. Instead of visiting every desk in the Marketing or Sales departments, the server assigns addresses automatically.

2.3.3.1. Configuring DHCP scope

A scope is the range of IP addresses that DHCP can lease. For XYZ Company, the scope covers the entire 10.0.0.0/8 network, and the first 50 addresses are excluded so they can be reserved for servers, printers, and network equipment.

1. Open the DHCP console from the Tools menu.
2. Expand XYZ-DC1, right-click IPv4, and select New Scope to start the wizard.
3. On the Scope Name screen, type XYZ_Clients and click Next.
4. In the IP Address Range screen, enter the following:
 - Start IP address: 10.0.0.1
 - End IP address: 10.255.255.254
 - Length: 8
 - Subnet mask: 255.0.0.0
5. Click Next. In the Add Exclusions and Delay screen, we must reserve the first 50 addresses for our infrastructure:
 - Start IP address: 10.0.0.1
 - End IP address: 10.0.0.50
6. Click Add, then click Next.
7. Set the Lease Duration to the default 8 days and click Next.
8. In the Configure DHCP Options screen, select Yes, I want to configure these options now and click Next.
9. For the Router (Default Gateway), type 10.0.0.1 and click Add, then Next.
10. In the Domain Name and DNS Servers screen, verify xyz.local is the parent domain and 10.0.0.1 is listed as the DNS server. Click Next.
11. Skip the WINS Server screen by clicking Next, then select Yes, I want to activate this scope now.
12. Click Finish.

New Scope Wizard

IP Address Range

You define the scope address range by identifying a set of consecutive IP addresses.



Configuration settings for DHCP Server

Enter the range of addresses that the scope distributes.

Start IP address:

End IP address:

Configuration settings that propagate to DHCP Client

Length:

Subnet mask:

< Back Next > Cancel

Figure: IP Address Range — 10.0.0.1 to 10.255.255.254 with /8 mask.

New Scope Wizard

Add Exclusions and Delay

Exclusions are addresses or a range of addresses that are not distributed by the server. A delay is the time duration by which the server will delay the transmission of a DHCP OFFER message.



Type the IP address range that you want to exclude. If you want to exclude a single address, type an address in Start IP address only.

Start IP address: End IP address:

Excluded address range:

10.0.0.1 to 10.0.0.50

Subnet delay in milli second:

< Back Next > Cancel

Figure: Exclusion 10.0.0.1 to 10.0.0.50 reserves the first 50 addresses for static use.

2.3.3.2. Configuring reservations

A reservation guarantees that a specific computer always receives the same IP address from DHCP, identified by its MAC address. In this case, we need to make two reservations: Finance-DB and IT-Fileserver. Reservations are created inside the scope, after the scope is activated.

1. In the DHCP console, expand the new XYZ_Clients scope.
2. Right-click Reservations and select New Reservation.
3. In the Reservation name field, type Finance-DB.
4. In the IP address field, type 10.0.0.10.
5. In the MAC address field, type 11-22-33-44-55-66.
6. Under Supported types, leave Both selected and click Add.
7. The form clears automatically for the next reservation. Repeat steps 3 to 6 for IT-Fileserver, using 10.0.0.11 as the IP and 33-44-55-66-77-88 as the MAC address.
8. Click Close and verify both reservations appear in the list.

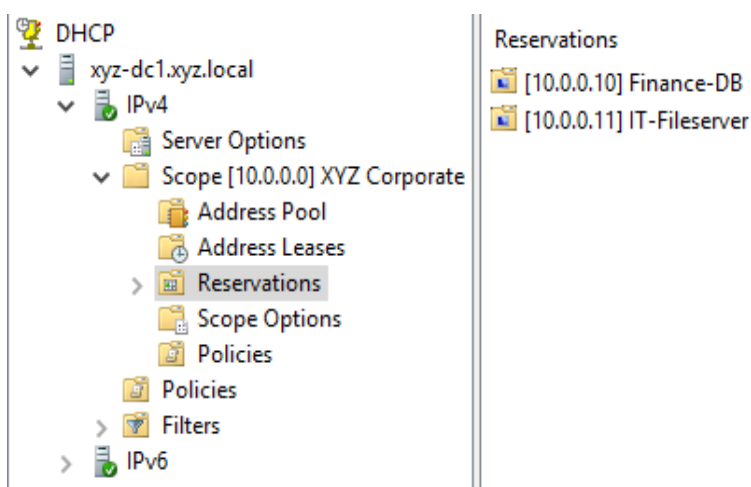


Figure: Reservations branch with Finance-DB (10.0.0.10) and IT-Fileserver (10.0.0.11).

2.4. Computers

In an Active Directory environment, computers must be joined to the domain to be managed by Group Policies. For the XYZ Company demonstration, we decided to pre-create the computer objects in their respective Organizational Units (OUs). This practice ensures that as soon as a physical workstation joins the domain, it immediately falls under the correct departmental policies without manual moving.

2.4.1. Creating Computers

To register the workstations in our database, we used the Active Directory Users and Computers console. We followed a naming convention based on the department and workstation number to keep the inventory organized.

1. Open Server Manager, click Tools, and select Active Directory Users and Computers.
2. Navigate through the xyz.local domain and locate the specific OU where the computer will reside (for example, the Marketing OU).
3. Right-click an empty space in the right pane, select New, and then click Computer.
4. In the Computer name field, type the designation for the machine. For this project we used MKTG-PC1 and MKTG-PC2 for Marketing, Finance-DB for Finance, and IT-Fileserver for Information Technology.
5. Verify that the Computer name (pre-Windows 2000) field is filled automatically with the same name.
6. Under the User or group section, you will see who is allowed to join this computer to the domain. By default, this is set to Domain Admins, which is what we used for our lab.
7. Click OK. The computer object will now appear in the OU with a computer icon, ready for the physical machine to be joined to the XYZ-DC1 controller.

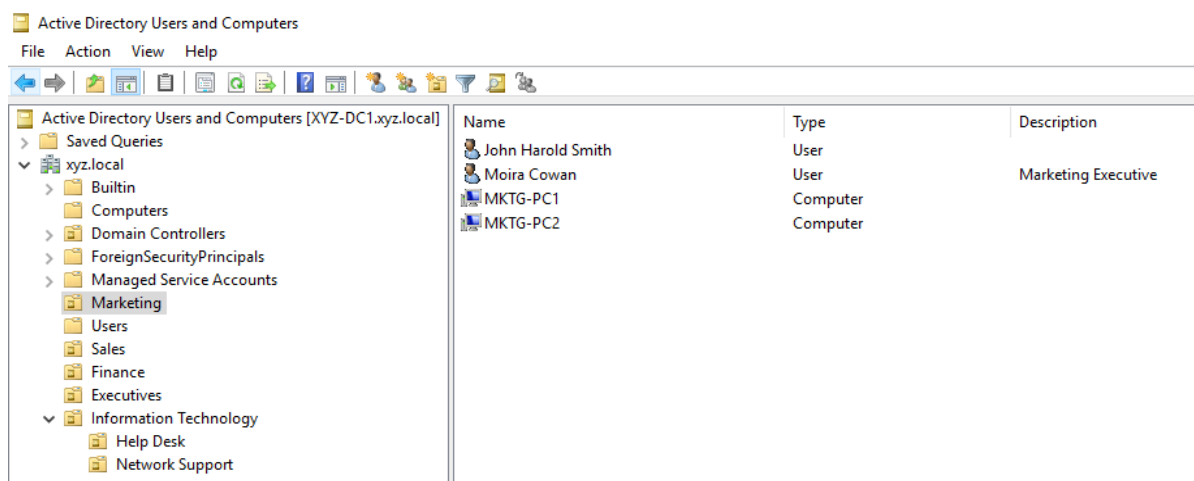


Figure: Marketing OU contains MKTG-PC1, MKTG-PC2, Moira Cowan, and John Harold Smith.

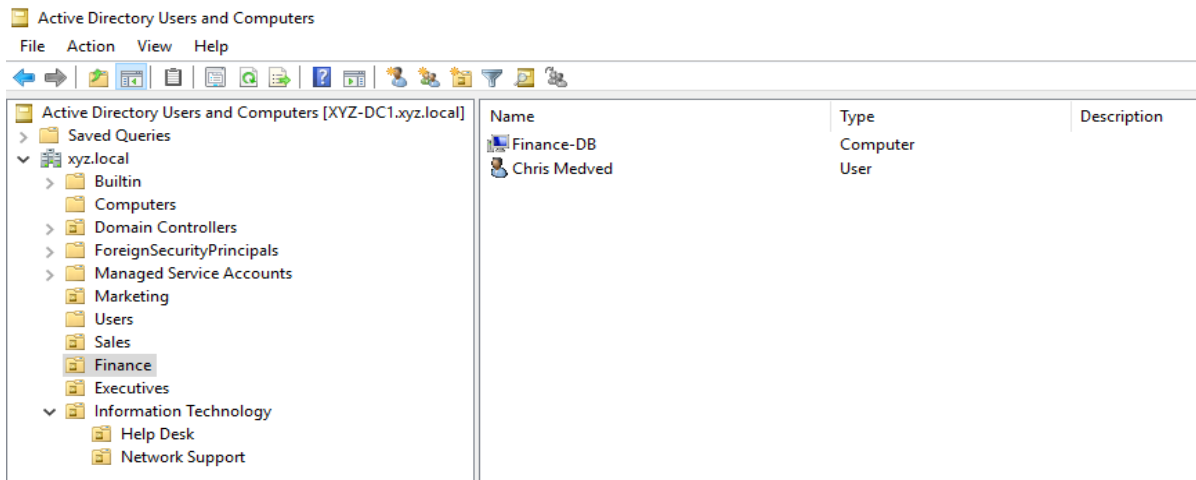


Figure: Finance OU contains Finance-DB and Chris Medved.

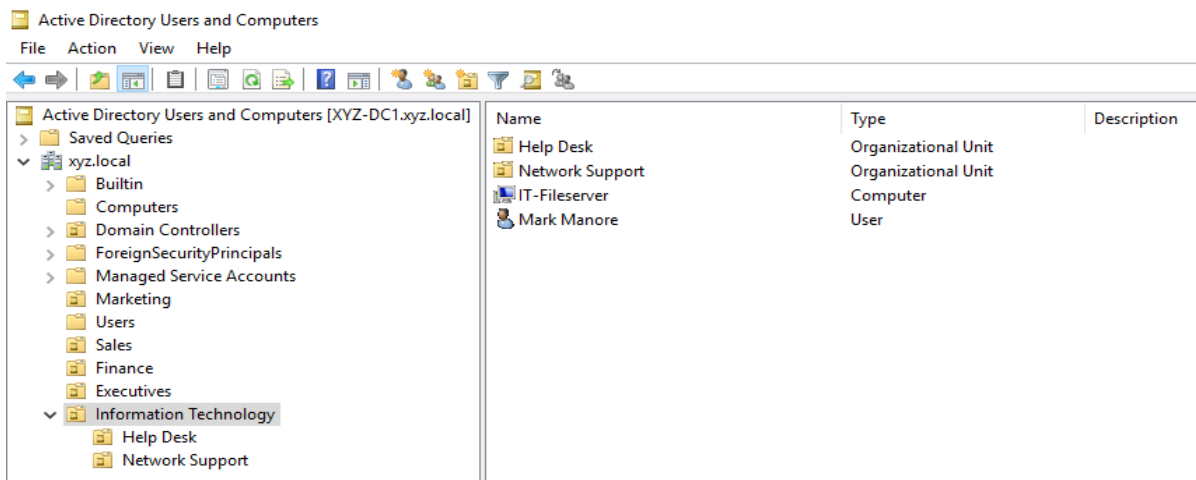


Figure: Information Technology OU with the sub-OUs Help Desk and Network Support, plus the IT-Fileserver computer and Mark Manore.

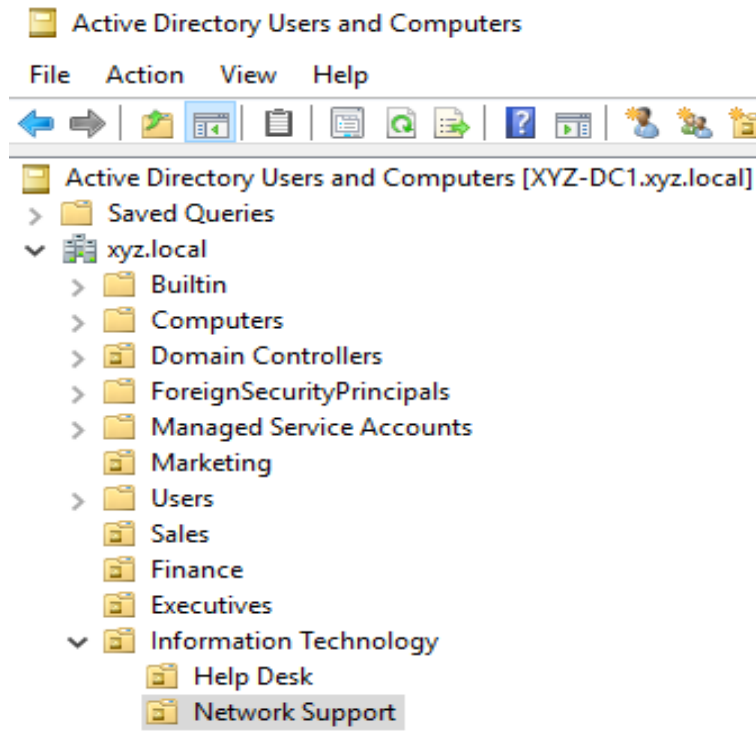


Figure: Complete OU hierarchy of the xyz.local domain.

2.5. Group Policy

Group Policy implementation is where we enforce the company's operating policies. For this project, we organized our resources into Organizational Units (OU's) to ensure that policies only affect the intended users and computers on XYZ-DC1.

2.5.1. Linking Policies

Before a policy can take effect, it must be linked to a specific container. In our lab, we linked department-specific rules to their respective OUs.

1. From the desktop, open Server Manager > Tools > Group Policy Management.
2. Expand Forest: xyz.local > Domains > xyz.local.
3. Right-click the desired OU (for example, Marketing) and select Create a GPO in this domain and Link it here....
4. Name the GPO Marketing Policy and click OK.
5. The new GPO will now appear linked to the OU, marked with a shortcut icon.

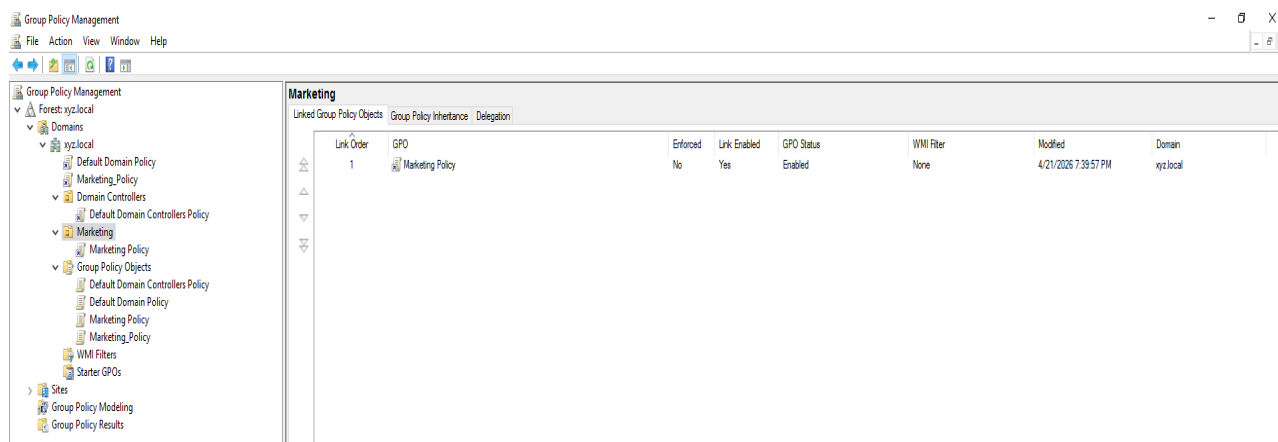


Figure: Marketing OU with Marketing Policy linked, Link Enabled = Yes, GPO Status = Enabled.

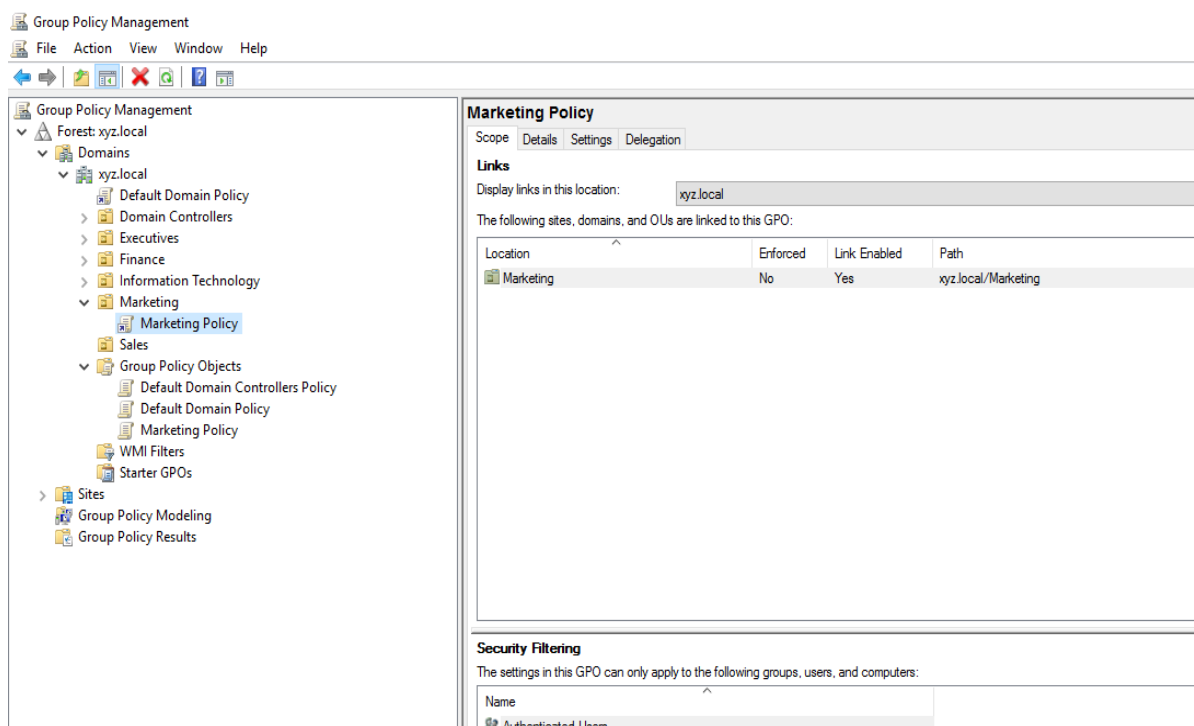


Figure: Marketing Policy — Scope tab confirms the link to xyz.local/Marketing.

2.5.2. Creating General Policies

These settings focus on the user's interface and environment restrictions.

1. In Group Policy Management, right-click your new GPO (Marketing_Policy) and select Edit.
2. To restrict the Control Panel, navigate to User Configuration > Policies > Administrative Templates > Control Panel.
3. Double-click Prohibit access to Control Panel and PC settings.
4. Select Enabled, then click Apply and OK.
5. To set the departmental wallpaper, navigate to User Configuration > Policies > Administrative Templates > Desktop > Desktop.
6. Edit the Desktop Wallpaper setting, set it to Enabled, and provide the local or network path to the image file.

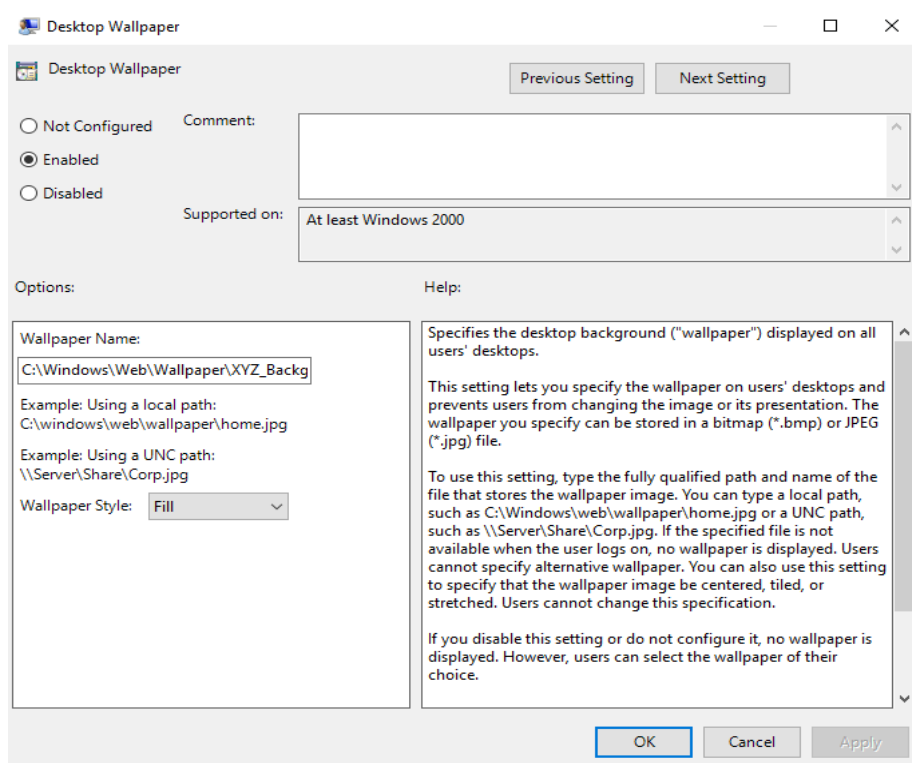


Figure: Desktop Wallpaper — Enabled, full path, Fill style.

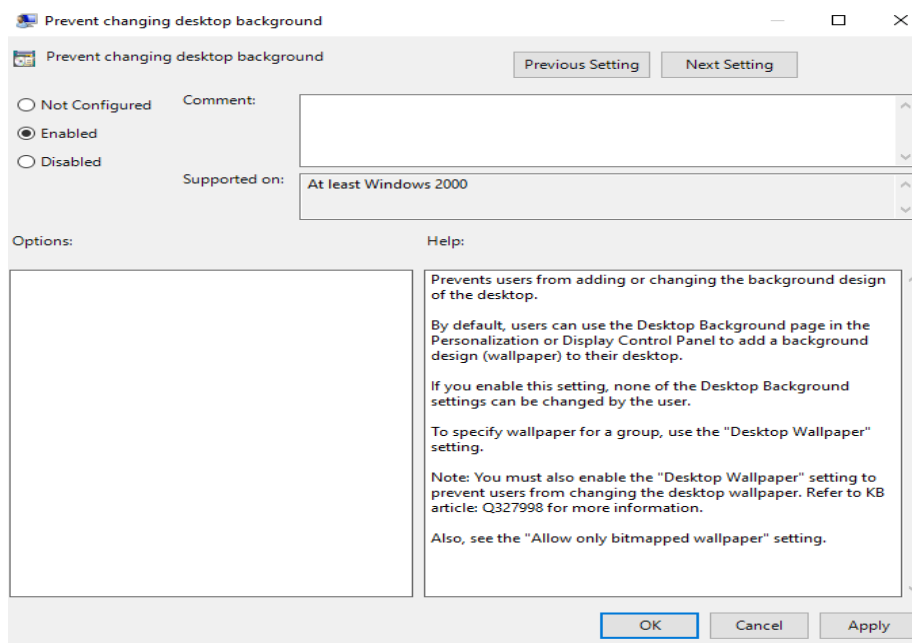


Figure: Prevent changing desktop background — Enabled.

We also used the Administrative Templates to further customize the user experience and improve security by removing unnecessary icons and menus.

1. In the same Desktop folder within Administrative Templates, locate and double-click Remove Recycle Bin icon from desktop. Select Enabled and click OK.
2. Navigate to User Configuration > Policies > Administrative Templates > Start Menu and Taskbar.
3. Double-click Remove Documents icon from Start Menu, select Enabled, and click OK.
4. In the same folder, locate Remove Run menu from Start Menu. Set this to Enabled to prevent users from executing unauthorized commands.
5. To keep the interface clean, double-click Remove user name from Start Menu, select Enabled, and click Apply.

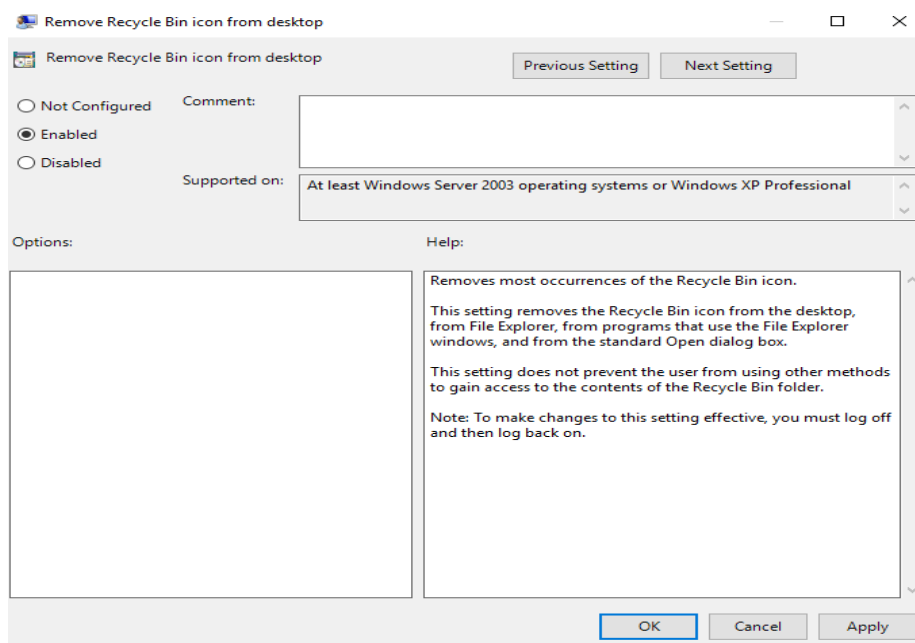


Figure: Remove Recycle Bin icon from desktop — Enabled.

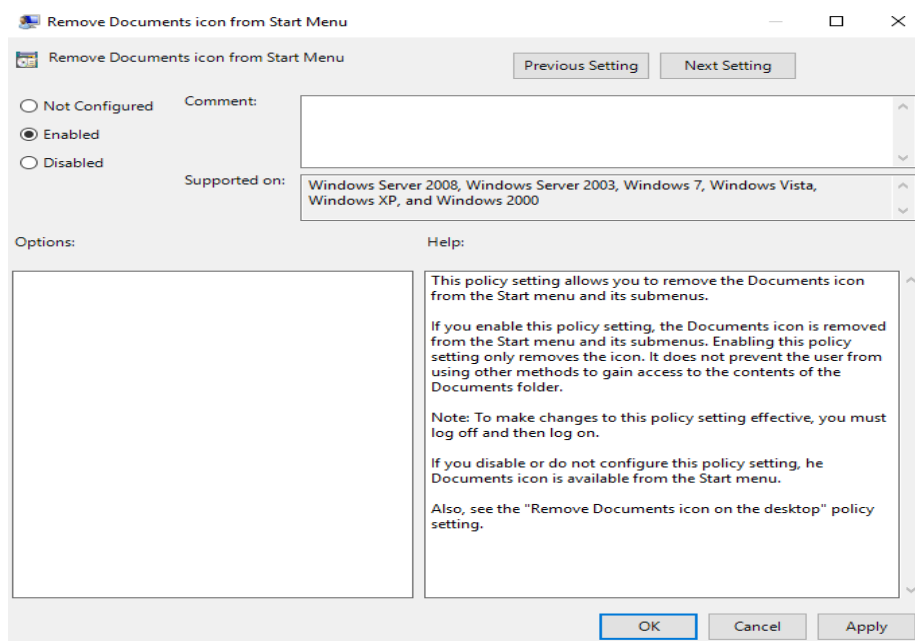


Figure: Remove Documents icon from Start Menu — Enabled.

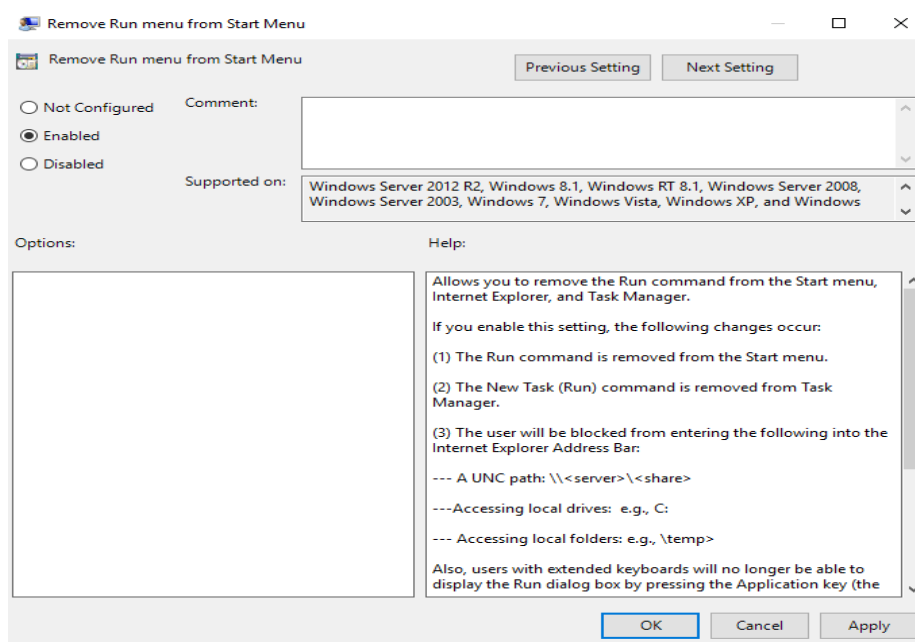


Figure: Remove Run menu from Start Menu — Enabled.

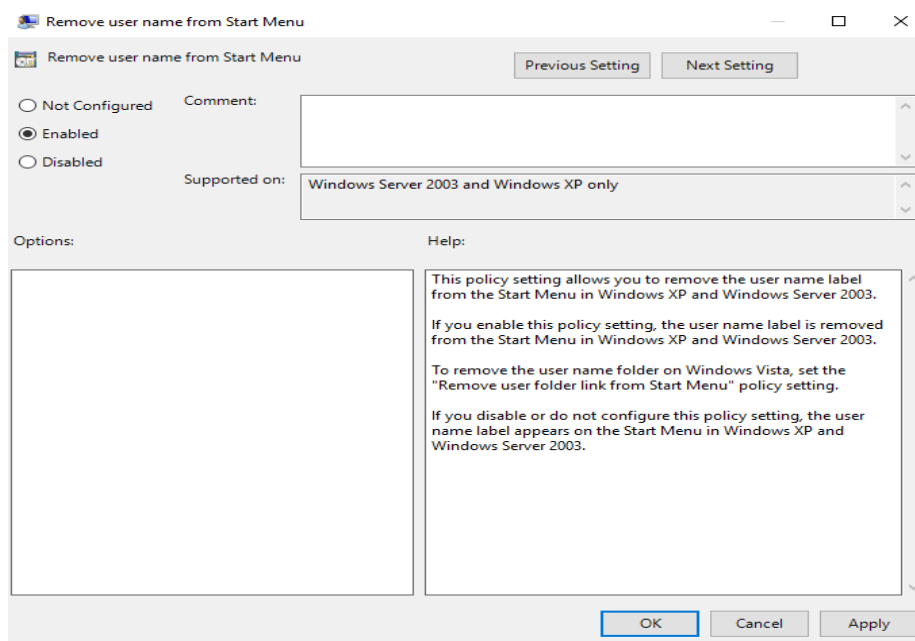


Figure: Remove user name from Start Menu — Enabled.

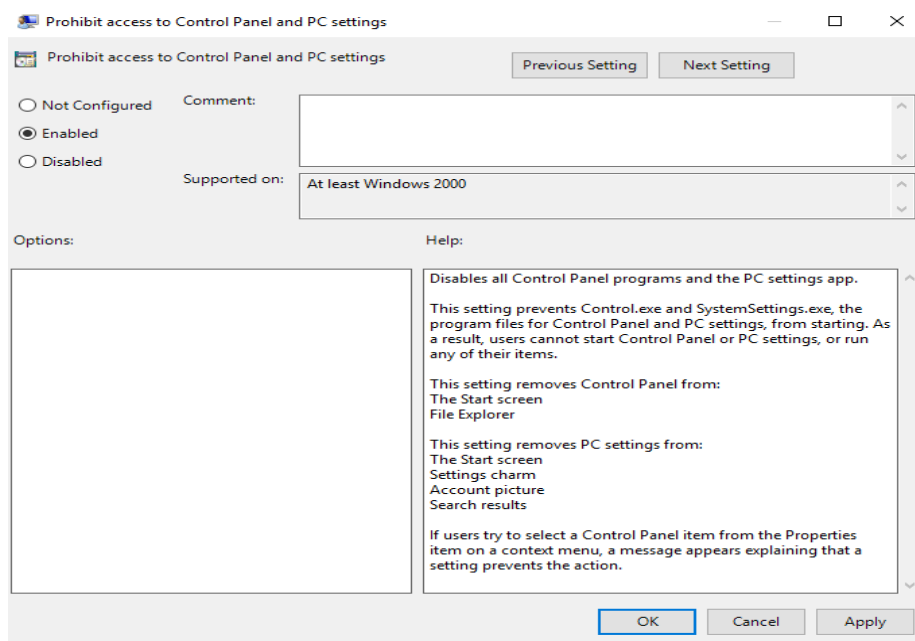


Figure: Prohibit access to Control Panel and PC settings — Enabled.

2.5.3. Creating Security Policies

Security policies protect the domain by enforcing rules like password complexity and login behavior. We also configured lockout restrictions to prevent unauthorized access through multiple failed login attempts.

1. In the GPO Editor, navigate to Computer Configuration > Policies > Windows Settings > Security Settings.
2. Expand Account Policies > Password Policy.
3. Double-click Enforce password history, check Define this policy setting, type 12 in the value field, and click OK.
4. Double-click Minimum password length, check Define this policy setting, type 5, and click OK.
5. Double-click Password must meet complexity requirements, check Define this policy setting, select Enabled, and click OK.

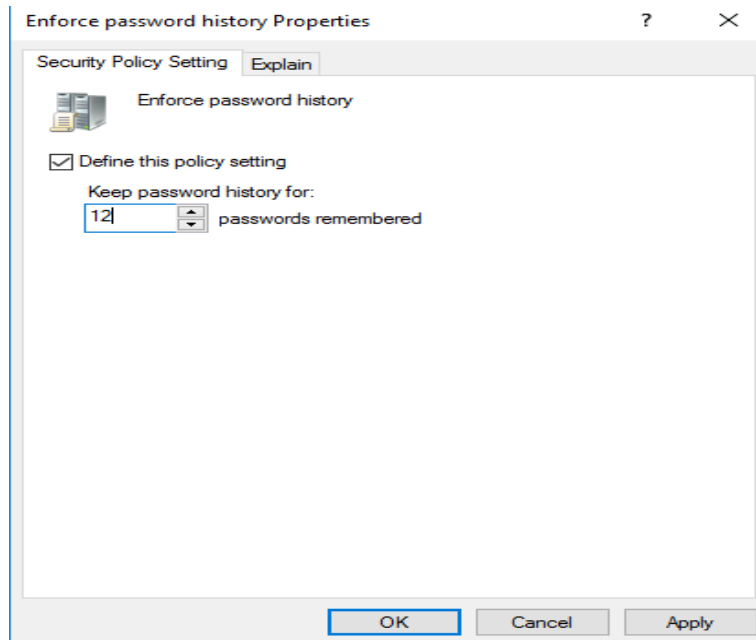


Figure: Enforce password history — 12 passwords remembered.

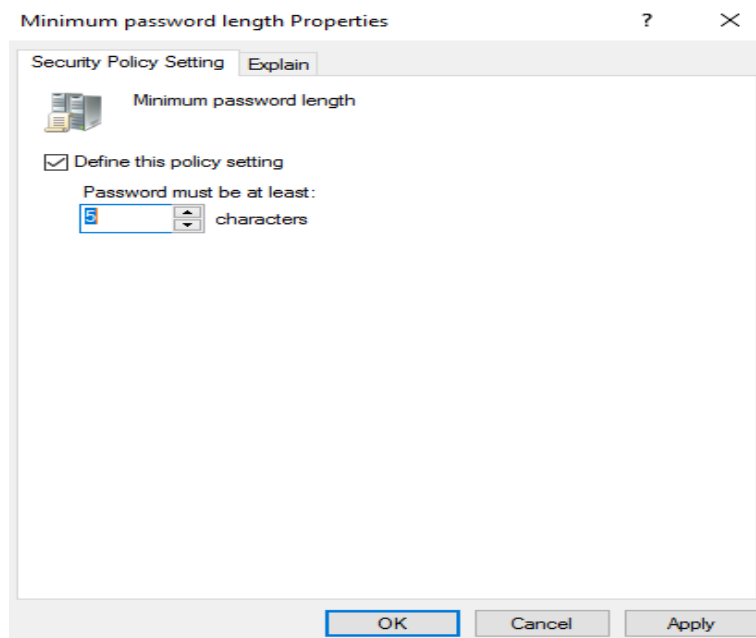


Figure: Minimum password length — 5 characters.

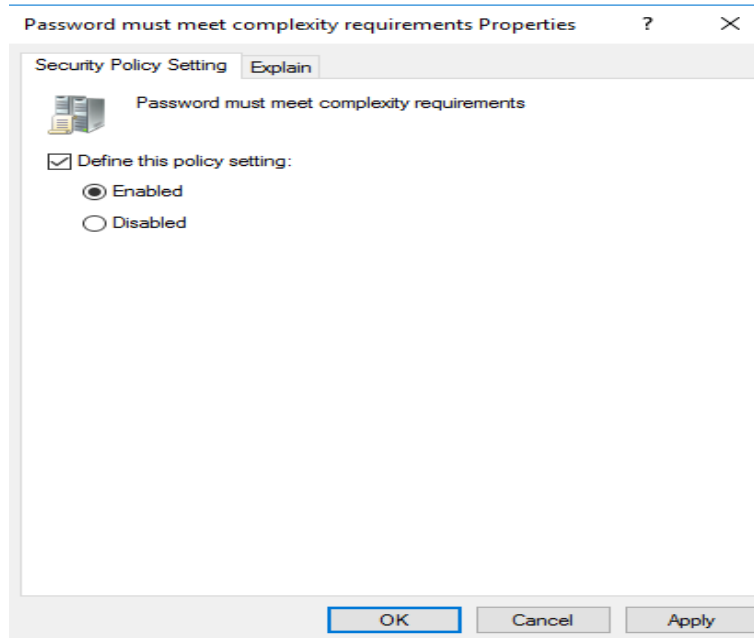


Figure: Password must meet complexity requirements — Enabled.

Account Lockout Policy Configuration:

To secure the XYZ Company against brute-force attacks, we established strict lockout thresholds.

1. In the same Account Policies branch, select the Account Lockout Policy folder.
2. Double-click Account lockout threshold. Set the value to 3 invalid logon attempts and click OK.
3. A dialog box will appear suggesting default values for the duration and the reset counter. We adjusted these according to the project requirements:
 - Set the Account lockout duration to 15 minutes.
 - Set the Reset account lockout counter after to 5 minutes.
4. Click Apply and OK to commit the security changes to the GPO.

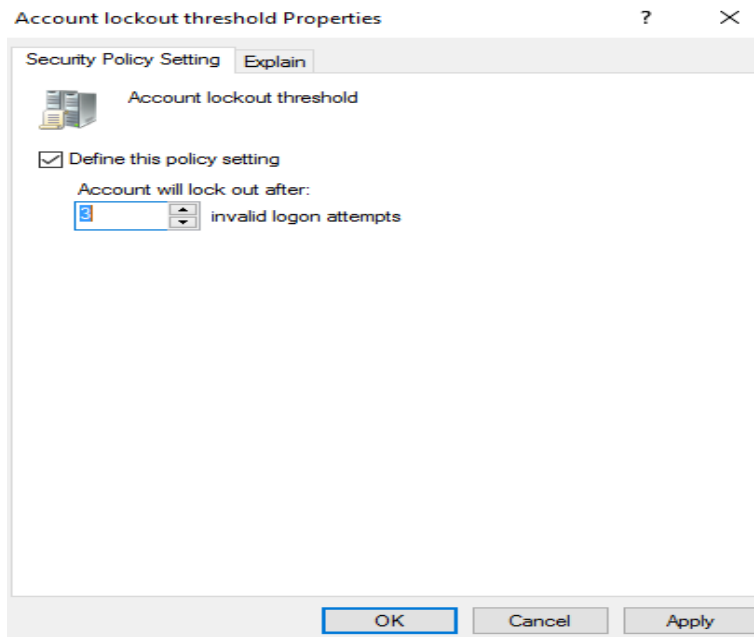


Figure: Account lockout threshold — 3 invalid logon attempts.

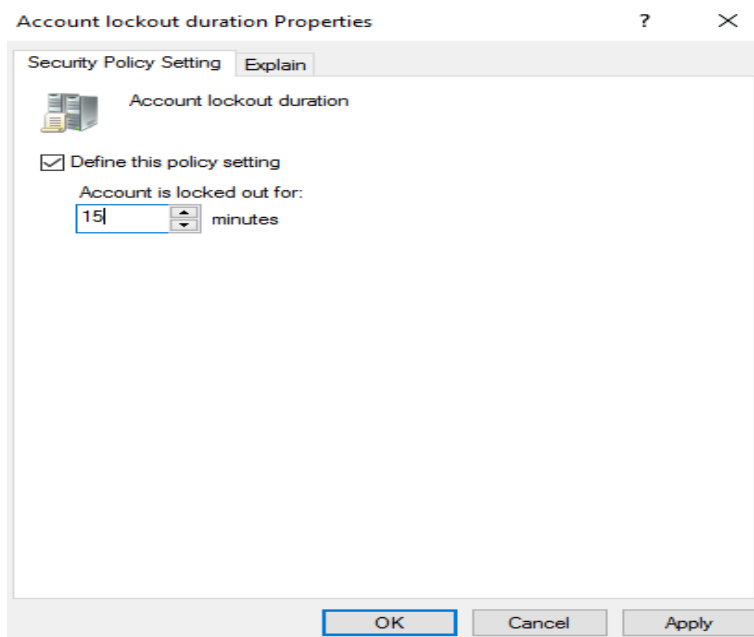


Figure: Account lockout duration — 15 minutes.

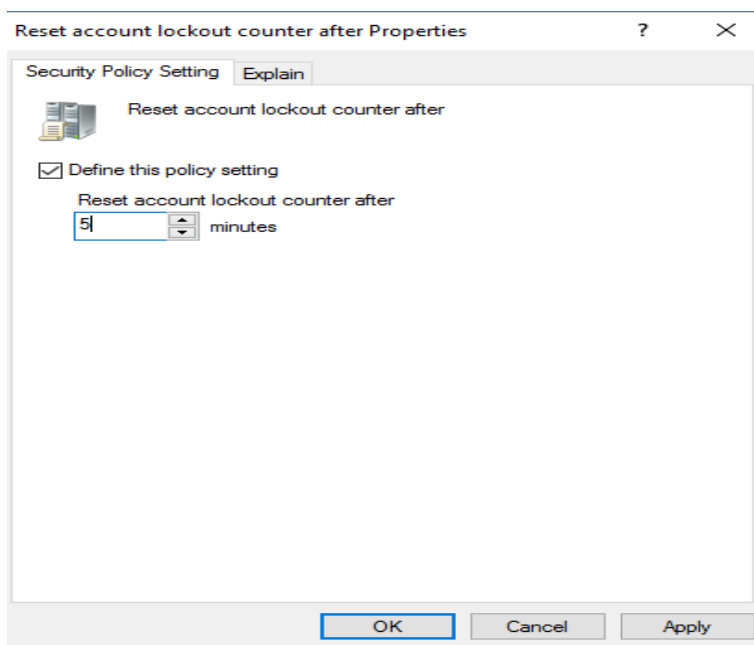


Figure: Reset account lockout counter after — 5 minutes.

Interactive Logon Banner Configuration:

To comply with the XYZ Company Security Policy, we configured a legal notice that appears every time a user attempts to log on to a workstation. This ensures all employees are aware of the system's usage terms.

1. In the same Security Options folder (located under Local Policies), find the policy titled Interactive logon: Message text for users attempting to log on.
2. Double-click the policy and check the box for Define this policy setting.
3. In the text box provided, type the exact text required by the project specification: Only Authorized Users of XYZ are permitted to log on. Click OK.
4. Additionally, locate the policy Interactive logon: Message title for users attempting to log on.
5. Enable it and type a title such as Security Notice or Authorized Access Only to ensure the banner is presented clearly.
6. Click Apply and OK.

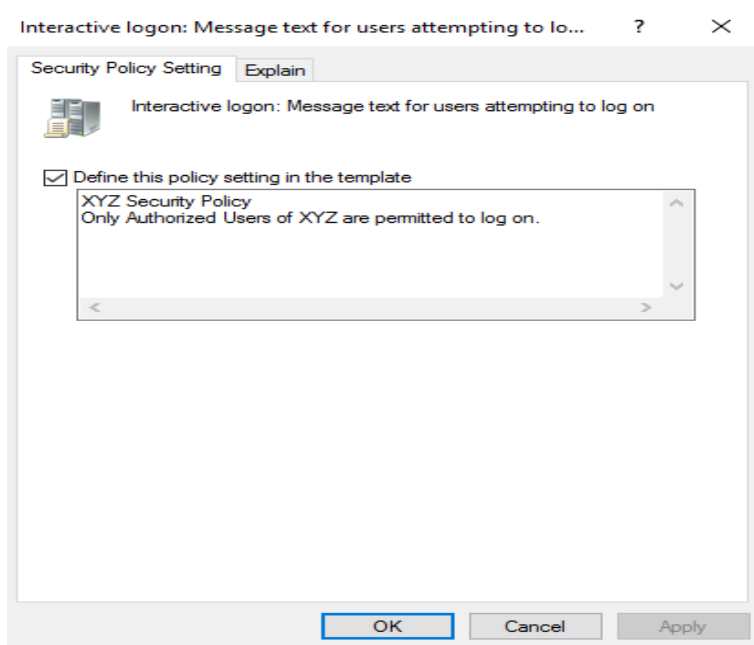


Figure: Interactive logon message — XYZ Security Policy.

2.5.4. Managing File System Permissions

The Reports folder lives at C:\Reports on the Domain Controller and is shared across the network so Marketing users can access it. Permissions are granted at two levels. Share permissions control access through the network. NTFS permissions control access through the file system itself and are more granular. The effective permission a user receives is the most restrictive of the two.

1. From the desktop, open File Explorer and create a new folder named Reports at the root of the C: drive.
2. Right-click the Reports folder and select Properties.
3. Open the Sharing tab and click Advanced Sharing.
4. Check the box Share this folder. The share name defaults to Reports. Click Permissions.
5. Click Add. Type Marketing Users in the object names field. Click Check Names, then OK. With Marketing Users selected, check the Read permission in the Allow column. Click OK twice to close the Sharing dialogs.
6. Back on the Reports Properties window, open the Security tab and click Edit.
7. Click Add. Type Marketing Users in the object names field. Click Check Names, then OK.
8. With Marketing Users selected, check the boxes for Read & execute, List folder contents, and Read in the Allow column. Click OK twice to apply the NTFS permissions.

- The shared folder is now reachable from any client computer in the domain through the path \\XYZ-DC1\\Reports, and only members of Marketing Users can read its contents.

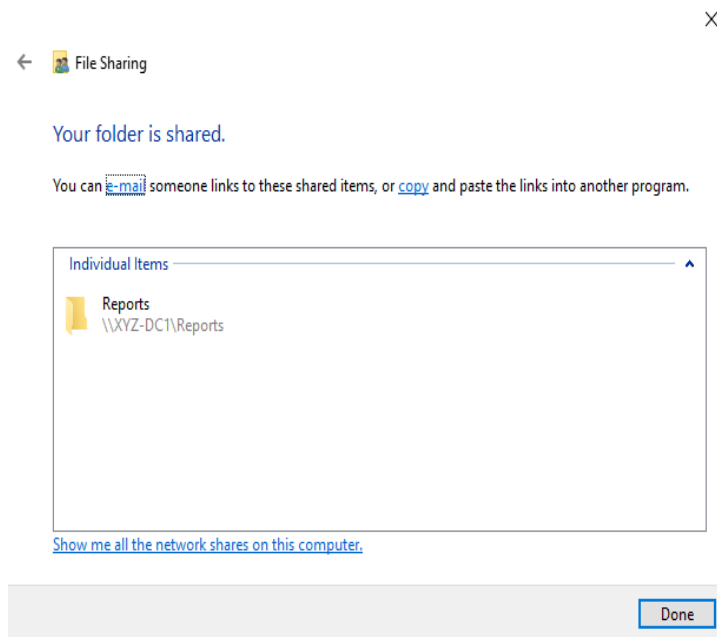


Figure: File Sharing dialog — Reports folder shared as \\XYZ-DC1\\Reports.

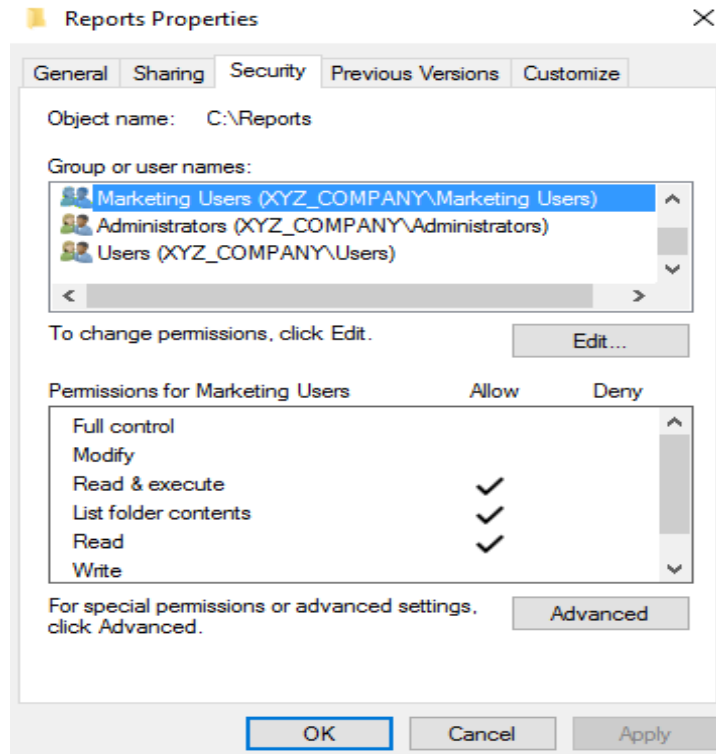


Figure: Security tab on C:\Reports — Marketing Users with Read & execute, List, Read.

2.5.5. Software Deployment

Finally, we use GPOs to automate the installation of essential tools like 7-Zip across the XYZ Company.

1. Ensure the 7-Zip.msi file is in a shared network folder where "Everyone" has Read access.
2. In the GPO Editor, go to User Configuration > Policies > Software Settings.
3. Right-click Software installation, select New, and then Package.
4. Type the full network path (UNC) to the file, such as \\XYZ-DC1\Software\7z2409-x64.msi, and click Open.
5. Select Assigned and click OK. Since this deployment is under User Configuration, 7-Zip will be installed automatically the next time a Marketing user signs in to any workstation in the domain.

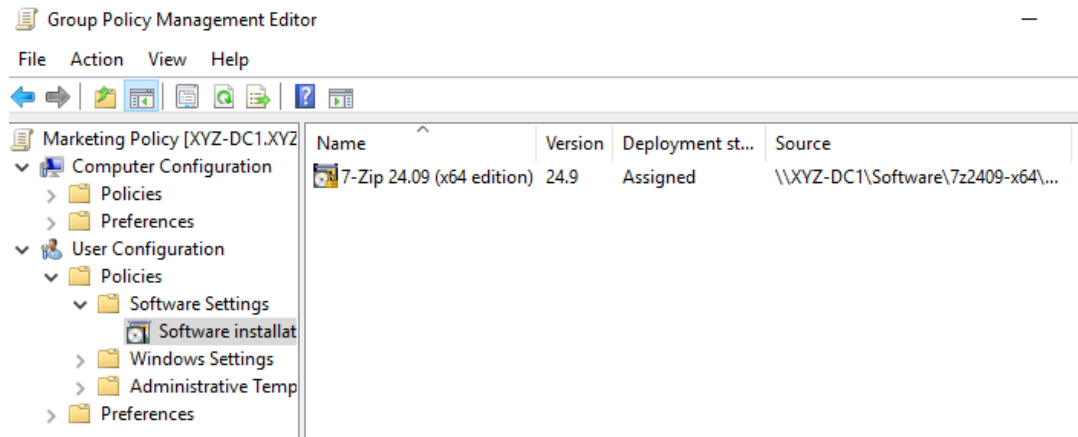


Figure: Software installation — 7-Zip 24.09 (x64 edition) Assigned, single deployment, sourced from \\XYZ-DC1\Software\7z2409-x64.msi.

2.6. Validating the Implementation

After completing the configuration of users, groups, and policies on XYZ-DC1, we must perform a series of tests to ensure the network is operational. Validation is the final step to confirm that the security settings and network services are being pushed correctly to the workstations.

2.6.1. Using GPUpdate and GPRresult

To avoid waiting for the automatic refresh cycle of the Active Directory, we manually forced the application of the Group Policy Objects (GPOs).

1. Open the Command Prompt with Administrator privileges.
2. Type `gpupdate /force` and press Enter. This ensures that both Computer and User policies are updated immediately.
3. Once the process is finished, type `gpresult /r` to view the Resultant Set of Policy (RSOP) report.
4. In the output, locate the Applied Group Policy Objects section. We verified that our specific policies, such as the Marketing Policy, were listed as successfully applied.

2.6.2. Verifying Network Services (DNS and DHCP)

We conducted tests to ensure that name resolution and IP addressing were functioning as intended for XYZ Company.

1. **DNS Test:** In the command prompt, we used the command `nslookup XYZ-DC1`. The server successfully resolved the name to the static IP 10.2.0.6, confirming the Forward Lookup Zone is healthy.
2. **DHCP Test:** We accessed the DHCP Console and checked the Address Leases folder. We confirmed that client machines were receiving dynamic IP addresses within the permitted range (starting from 10.0.0.51), effectively skipping the exclusion range we set earlier.

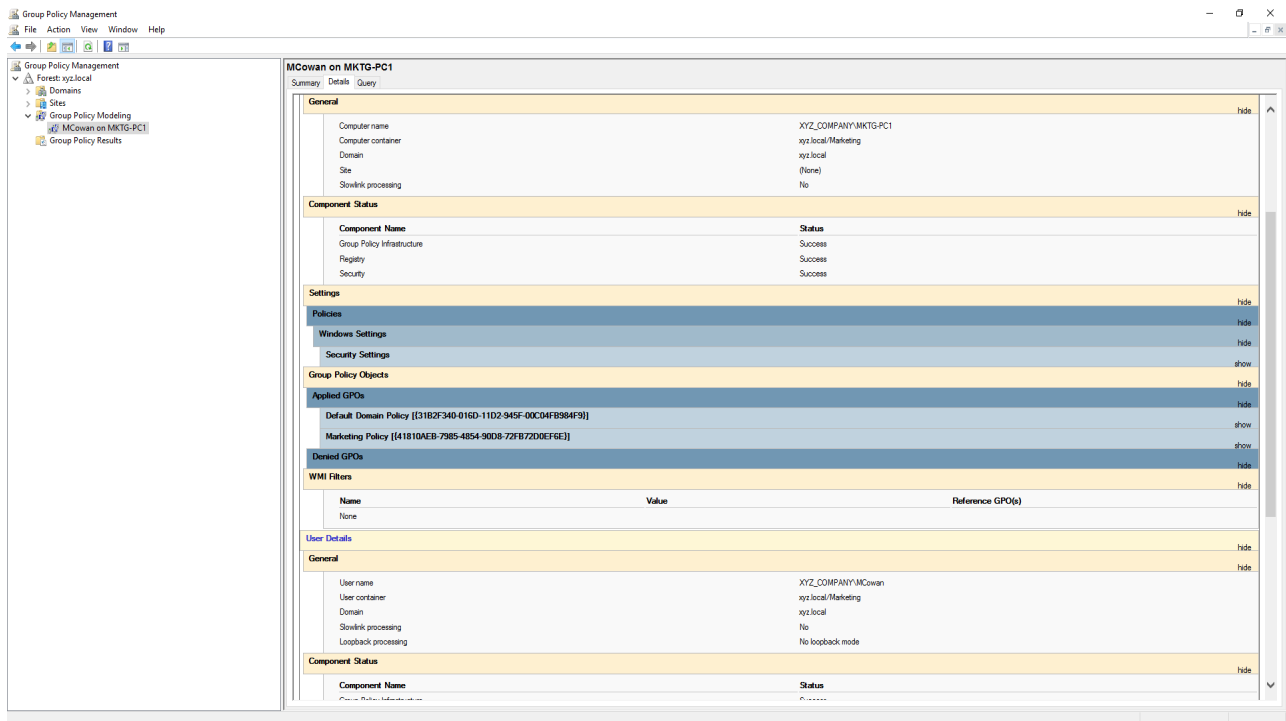


Figure: Modeling result for MCowan on MKTG-PC1 — Marketing Policy listed in Applied GPOs and all components show Success.

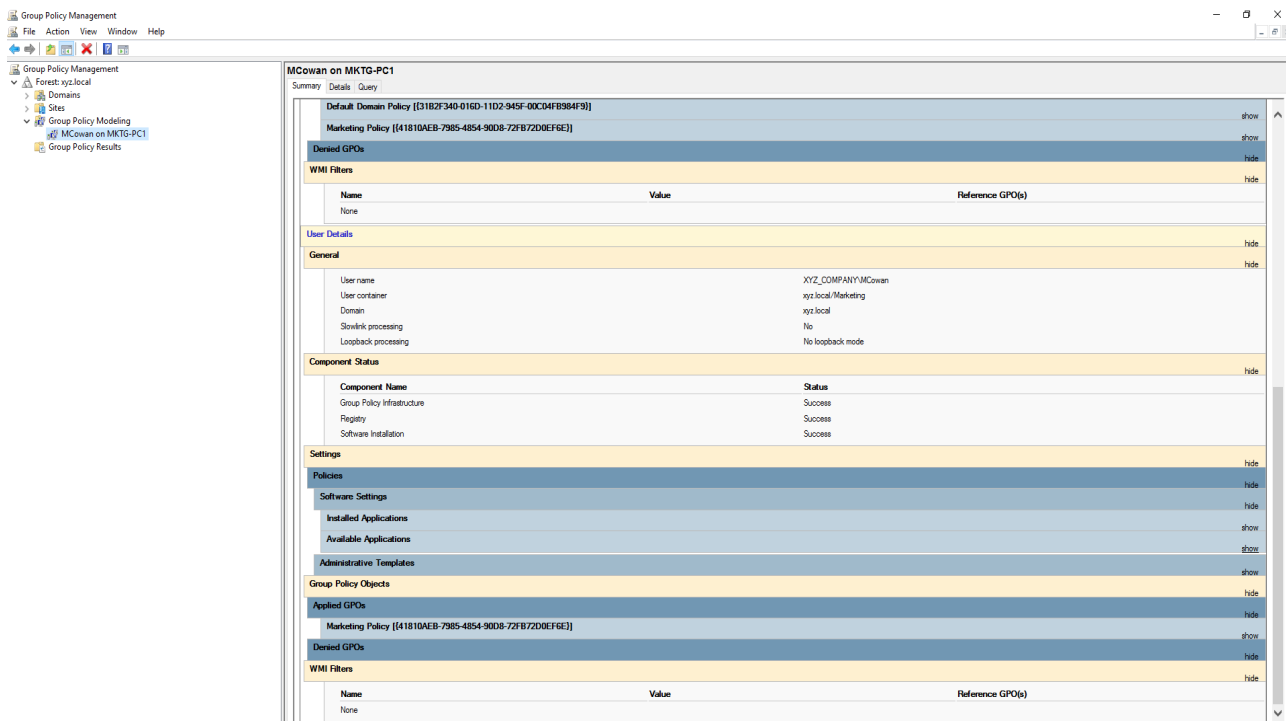


Figure: Modeling — Settings tab confirms Software Installation, Group Policy Objects, and Applied GPOs.

Marketing Policy

Scope

Details

Settings

Delegation

Marketing Policy

Data collected on: 4/28/2026 8:39:56 PM

show all

General

hide

Details

hide

Domain

xyz.local

Owner

XYZ_COMPANY\Domain Admins

Created

4/14/2026 6:49:36 PM

Modified

4/28/2026 8:20:52 PM

User Revisions

14 (AD), 14 (SYSVOL)

Computer Revisions

4 (AD), 4 (SYSVOL)

Unique ID

{41810AEB-7985-4854-90D8-72FB72D0EF6E}

GPO Status

Enabled

Links

hide

Location

Enforced

Link Status

Path

Marketing

No

Enabled

xyz.local/Marketing

This list only includes links in the domain of the GPO.

Security Filtering

hide

The settings in this GPO can only apply to the following groups, users, and computers:

Name

NT AUTHORITY\Authenticated Users

Delegation

hide

These groups and users have the specified permission for this GPO

Name

Allowed Permissions

Inherited

NT AUTHORITY\Authenticated Users

Read (from Security Filtering)

No

NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS

Read

No

NT AUTHORITY\SYSTEM

Edit settings, delete, modify security

No

XYZ_COMPANY\Domain Admins

Edit settings, delete, modify security

No

XYZ_COMPANY\Enterprise Admins

Edit settings, delete, modify security

No

Computer Configuration (Enabled)

hide

Policies

hide

Windows Settings

hide

Security Settings

hide

Account Policies/Password Policy

hide

Figure: Marketing Policy Settings report — General details, Links, Security Filtering, and Delegation.

Marketing Policy	
Scope	Details Settings Delegation
Account Policies/Password Policy	
Policy	Setting
Enforce password history	12 passwords remembered
Minimum password length	5 characters
Account Policies/Account Lockout Policy	
Policy	Setting
Account lockout duration	15 minutes
Account lockout threshold	3 invalid logon attempts
Reset account lockout counter after	5 minutes
Local Policies/Security Options	
Interactive Logon	
Policy	Setting
Interactive logon: Message text for users attempting to log on	XYZ Security Policy. Only Authorized Users of XYZ are permitted to log on.
User Configuration (Enabled)	
Policies	
Software Settings	
Assigned Applications	
7-Zip 24.09 (x64 edition)	
Product Information	
Name	7-Zip 24.09 (x64 edition)
Version	24.09
Language	English (United States)
Platform	x64
Support URL	http://www.7-zip.org/support.html
Deployment Information	
General	Setting
Deployment type	Assigned
Deployment source	\\XYZ-DC1\Software\7z2409x64\7z2409x64.msi
Installation user interface options	Maximum
Uninstall this application when it falls out of the scope of management	Disabled
Do not display this package in the Add/Remove Programs control panel	Disabled
Install this application at logon	Disabled
Advanced Deployment Options	
Setting	

Figure: Marketing Policy Settings report — Account Policies, Local Policies, and Software Settings.

Marketing Policy

Scope Details Settings Delegation

Upgrades		Setting
Required upgrade for existing packages		Disabled
Packages that this package will upgrade		GPO
None		
Packages in the current GPO that will upgrade this package		None
Categories		
None		
Transforms		
None		
Administrative Templates		
Policy definitions (ADMX files) retrieved from the local computer.		
Control Panel		
Policy	Setting	Comment
Prohibit access to Control Panel and PC settings	Enabled	
Control Panel/Personalization		
Policy	Setting	Comment
Prevent changing desktop background	Enabled	
Desktop		
Policy	Setting	Comment
Remove Recycle Bin icon from desktop	Enabled	
Desktop/Desktop		
Policy	Setting	Comment
Desktop Wallpaper	Enabled	
Wallpaper Name:		C:\Windows\Web\Wallpaper\XYZ_Background.jpg
Example: Using a local path: C:\windows\web\wallpaper\home.jpg		
Example: Using a UNC path: \\Server\Share\Corp.jpg		
Wallpaper Style:		Fill
Start Menu and Taskbar		
Policy	Setting	Comment
Remove Documents icon from Start Menu	Enabled	
Remove Run menu from Start Menu	Enabled	
Remove user name from Start Menu	Enabled	

Figure: Marketing Policy Settings report — Administrative Templates including Desktop Wallpaper.

Confirmation of the GPO refresh is captured at the Administrator command prompt:

```
Administrator: Command Prompt

Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.
```

Figure: gpupdate /force — Computer Policy and User Policy updated successfully.

3. References

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